THE CARBOX MONOGRAPH & MASTER REFERENCE MANUAL

Structural Identification of Anomalous Carboxyl Protonation States, Low-Barrier Hydrogen Bonds, and Microenvironment Dielectric Shielding in High-Resolution Protein Crystal Structures

A Comprehensive Treatise on Macromolecular Acid-Base Equilibria, Crystallographic Coordinate Integrity, Real-Space Difference Density Auditing, and Automated Refinement Restraint Integration

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The Carbox Crystallographic Validation Axiom

«■g■r atomlar üçün elektron paylanmas■ yoxdursa onda onlar s■hv modell■■dirilib. ■g■r atomlar s■hv modell■■dirilibs■ onda onlar aras■nda m■saf■ m■nas■zd■r.» (Garib N. Murshudov: «If there is no electron density distribution for the atoms, then they are modeled incorrectly. If the atoms are modeled incorrectly, then the distance between them is meaningless.»)

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Architecture & Engineering Credits:
Carbox and CarboxMol are developed and maintained by Suleiman M Hajizadeh and Garib N Murshudov, providing native high-throughput crystallographic analysis, empirical pKa prediction, and 3D molecular modeling.

Scientific Warning on Crystallographic Distance Mining

Spatial proximity alone (e.g. d(O...O) <= 3.20 A) is insufficient to infer Low-Barrier Hydrogen Bonds (LBHB). In naive PDB screenings, over 90% of apparent ultra-short carboxyl contacts arise from high thermal motion (B-factor outliers), refinement steric clashes, unmodeled alternate conformations, or unmodeled divalent cations. The Carbox Platform Validation Engine must always be executed to verify real-space electron density.

Foreword by Prof. Garib N. Murshudov

For over three decades, macromolecular crystallography has driven biological discovery by providing atomic-resolution models of enzymes, receptors, and complex supramolecular assemblies. Yet, as our refinement methods have matured, an insidious trap has persisted in automated data mining: the confusion of coordinate numbers with physical reality. When an X-ray structure is deposited in the Protein Data Bank with an inter-atomic distance of 2.40 Å between two carboxylate oxygens, bioinformaticians frequently rush to claim the discovery of a Low-Barrier Hydrogen Bond or an unusual enzymatic state.

However, crystallography is an experimental science of electron density scattering, not a geometric game of rigid spheres. If a residue sits in a flexible surface loop where the electron density is faint or non-existent, the crystallographic refinement program must still place atoms somewhere. If the stereochemical restraints are loose, or if water molecules were omitted, atoms can easily be pushed into mutual steric clash. To claim that such atoms form an ultra-strong hydrogen bond is to misinterpret thermal disorder as chemical bonding.

In Carbox, Suleiman M Hajizadeh and I have sought to establish an uncompromising, scientifically sound platform that bridges theoretical biophysics, high-throughput bioinformatic mining, and rigorous crystallographic validation. By establishing statistical B-factor filtering, standardized Z-scores, real-space difference density verification, and direct integration with modern refinement architectures, Carbox ensures that researchers only build hypotheses upon coordinates that are genuinely supported by experimental observations.

Prof. Garib N. Murshudov
MRC Laboratory of Molecular Biology, Cambridge, UK

Preface by Suleiman M Hajizadeh

The inception of Carbox arose from a simple observation: while carboxyl groups are central to almost every catalytic mechanism in biochemistry, structural biologists had no unified, accessible instrument to evaluate their protonation states, inspect their density contours, and generate correct refinement parameters. Desktop software was fragmented, command-line pipelines were difficult to configure, and web viewers lacked the precision required for crystallographic investigation.

Carbox v0.4.5 represents the realization of a complete ecosystem. From a zero-dependency Python graph engine to a zero-install trilingual WebGL workbench featuring CarboxMol precision gesture controls, Carbox democratizes advanced structural biology. This manual has been composed as an exhaustive reference text for students, structural bioinformaticians, and crystallographers alike.

Suleiman M Hajizadeh
Baku, Azerbaijan

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Macromolecular Crystallography, Quantum Biophysics of Carboxylate–Carboxylate Dyads, High-Throughput Protonation State Modeling, Electron Density Validation, and 3D Graphical Workbench Architecture

Authors & Scientific Direction

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Repository: https://github.com/project-carbox/carbox

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CHAPTER SECTION

1. PROLEGOMENA & EXECUTIVE SUMMARY

The **Carbox & CarboxMol Project** represents an exhaustive, rigorous computational biophysics and macromolecular crystallography suite dedicated to the discovery, validation, simulation, and refinement of anomalous carboxylate–carboxylate interactions in biological macromolecules.

For decades, structural biologists operating the Protein Data Bank (PDB) have confronted a perplexing phenomenon: hundreds of crystallographic models report pairs of acidic sidechains—Aspartate (Asp) and Glutamate (Glu)—whose carboxylate oxygen atoms reside at interatomic distances well below the sum of their van der Waals radii ($d(O\cdots O) < 3.20$ Å), in some cases reaching extreme distances of $2.20 - 2.50$ Å. Under classical physiological conditions at neutral pH (~7.4), both carboxyl groups are expected to be fully deprotonated and carry a formal negative charge ($-1e$). Bringing two unshielded negative charges to within 2.40 Å incurs an immense electrostatic repulsion energy exceeding $+100\text{ kcal/mol}$.

Historically, two diametrically opposed interpretations arose in structural biology:

- **The Biological / Quantum Viewpoint:** The two carboxylates share a single proton in an ultra-short, highly covalent **Low-Barrier Hydrogen Bond (LBHB)** or Short Strong Hydrogen Bond (SSHB), wherein the zero-point vibrational energy of the proton meets or exceeds the barrier between potential wells, stabilizing the active site and driving enzyme catalysis.
- **The Crystallographic / Validation Viewpoint (The Carbox Axiom):** A vast proportion of these reported contacts are crystallographic artifacts—products of model building errors, misassigned sidechain rotamers, unmodeled bridging water molecules or divalent metal cations, or residues situated in mobile, disordered surface loops where thermal vibrations render coordinates statistically meaningless.

The Carbox ecosystem was designed to bridge this divide through mathematical rigor, crystallographic evidence, and modern interactive graphics. Carbox delivers:

- An automated high-throughput scanning and graph analysis pipeline (**`carbox/`**) capable of surveying the entire 200,000+ entry Protein Data Bank.
- The **Carbox Platform Validation Engine**, establishing statistical thresholds over atomic displacement parameters (*B*-factors) and local electron density to immediately separate true catalytic dyads from mobile loop artifacts.
- A quantum potential energy surface (PES) simulator modeling the one-dimensional proton transfer coordinate between carboxyl oxygens.
- An automated crystallographic restraint generator generating external distance, angle, and **`MODRES`** protonation records for modern refinement programs (**Carbox Refinement**, **Carbox Refinement**, and **Phenix**).
- **CarboxMol Workbench**, a full-featured desktop crystallography workbench dedicated to high-resolution structural analysis, difference density visualization, and restraints modeling.
- **Carbox Web Studio**, a state-of-the-art, zero-dependency browser workbench powered by CarboxMol WebGL, featuring real-time Mode 0 CCP4 map decompression, 1D density profiling, trilingual localization (English, Azerbaijani, Hebrew), and a CarboxMol precision gesture engine supporting Mac trackpads and discrete mouse wheels.

This monograph serves as the definitive reference manual for the entire codebase, its underlying physical theories, and its operational mechanics.

CHAPTER SECTION

2. BIOPHYSICAL FOUNDATIONS: THE CARBOXYLATE–CARBOXYLATE MICROENVIRONMENT

2.1 The Acidic Amino Acid Sidechains: Aspartate and Glutamate

Aspartate (Asp, D) and Glutamate (Glu, E) possess sidechain carboxyl groups (-*text{COOH} *rightleftharpoons -*text{COO}^- + *text{H}^+). In dilute aqueous solution, their intrinsic acid dissociation constants (*text{p}K_a) are:

$*text{p}K_a(*text{Asp, free}) *approx 3.90 *pm 0.10$

$*text{p}K_a(*text{Glu, free}) *approx 4.25 *pm 0.10$

At cytosolic pH (*7.2 - 7.4*), the Henderson-Hasselbalch equation indicates that more than *99.9\%* of free Asp and Glu residues exist in the deprotonated, resonance-stabilized carboxylate form:

$*text{pH} = *text{p}K_a + *log_{10}*left(*frac{[-*text{COO}^-]}{[-*text{COOH}]}*right) *implies *frac{[-*text{COO}^-]}{[-*text{COOH}]} = 10^{7.4 - 4.0} *approx 2500$

In the resonance structure of the carboxylate anion, the negative charge is distributed equally across both oxygen atoms (*O*delta1, O*delta2* in Asp; *O*epsilon1, O*epsilon2* in Glu), with bond orders of approximately *1.5* and typical carbon–oxygen bond lengths of *1.25* Å. In the neutral carboxylic acid form (-*text{COOH}), the symmetry is broken: the carbonyl oxygen (*C=O*) exhibits a bond length of *sim *1.21* Å, while the hydroxyl oxygen (*C-OH*) exhibits a length of *sim *1.31* Å.

2.2 The Paradox of Ultra-Short Carboxyl–Carboxyl Contacts

The van der Waals radius of an sp² oxygen atom is universally accepted as *r_{vdW} *approx 1.52* Å. Therefore, the non-bonded contact distance below which severe steric overlap occurs is:

$*d_{sum_vdW} = 1.52*text{ *AA} + 1.52*text{ *AA} = 3.04*text{ *AA}$

When two carboxylate oxygens are brought to *d(O*cdots O) *le 2.60* Å, their electron clouds overlap significantly. If both carboxylates remain negatively charged, the Coulombic repulsion energy in a vacuum or low-dielectric environment is given by:

$*V_{*text{Coulomb}} = *frac{1}{4*pi*varepsilon_0 *varepsilon_r} *cdot *frac{q_1 q_2}{r_{12}} = *frac{332.063}{*varepsilon_r *cdot r_{12}*text{ [*AA]}}*text{ kcal/mol}$

For *q_1 = q_2 = -1e*, *r_{12} = 2.45* Å, and an interior protein core dielectric of **varepsilon_r *approx 4.0*:

$*V_{*text{Coulomb}} = *frac{332.063}{4.0 *times 2.45} *approx +33.9*text{ kcal/mol}$

In a hydrophobic hydrophobic pocket (**varepsilon_r *approx 2.0*), this repulsive barrier skyrockets to *+67.8*text{ kcal/mol}*. A folded protein structure cannot sustain such energetic penalties without immediate conformational disruption or active site unfolding unless compensating physical mechanisms exist.

2.3 Energetics of Electrostatic Repulsion vs. Hydrogen Bonding

To stabilize two carboxyl groups in close proximity, nature utilizes three distinct thermodynamic strategies:

- **Mono-Protonation:** One carboxylate picks up a proton, transforming the repulsive interaction into an attractive hydrogen bond (-*text{COO}^- *cdots *text{HOOC}-).
- **Divalent Metal Bridging:** A positively charged cation (e.g., *text{Mg}^{2+}, *text{Ca}^{2+}, *text{Zn}^{2+}) sits between the two carboxylates, coordinating both negative charges and turning repulsion into a highly stable coordination complex.
- **Dielectric Screening / Water Networks:** Multiple ordered water molecules form a bridging hydrogen-bonded network that disperses the charge.

When mono-protonation occurs, one of the residues experiences a massive *text{p}K_a shift. If the microenvironment forces one residue to remain protonated at neutral pH (**text{pH} = 7.4*), its apparent *text{p}K_a must be shifted upward by

3 to *6* units:

$\text{p}K_{a,\text{perturbed}} \ge 7.5 - 9.5$

2.4 Low-Barrier Hydrogen Bonds (LBHBs) vs. Standard Hydrogen Bonds

In standard biological hydrogen bonds (e.g., peptide \text{N}-\text{H}\cdots\text{O}=\text{C}), the distance between heavy atoms is *d(N\cdots O) \approx 2.80 - 3.10* Å. The proton remains localized in a deep potential well covalently attached to the donor atom, and the bond energy is modest:

$\Delta H_{\text{standard HB}} \approx -2\text{ to }-5\text{ kcal/mol}$

However, as the donor–acceptor distance compresses below *2.55* Å, the interaction changes qualitatively into a **Low-Barrier Hydrogen Bond (LBHB)** or **Short Strong Hydrogen Bond (SSHB)**:

- **Heavy atom distance:** *d(O\cdots O) \le 2.55* Å (in extreme cases, *2.38 - 2.48* Å).
- **Bond energy:** *\Delta H_{\text{LBHB}} \approx -12\text{ to }-25\text{ kcal/mol}*.
- **Covalency:** Significant electron density accumulation between the hydrogen and both oxygen atoms.
- **Proton delocalization:** The zero-point vibrational energy of the proton is comparable to or exceeds the central potential barrier.
- **Isotope effect:** Marked deviation from classical semi-classical fractionation factors (*D/H* fractionation factor *\phi \ll 1.0*).
- **NMR signature:** Unusually high chemical shifts (*\delta > 16 - 21\text{ ppm}*) for the bridging proton.

```
Standard Hydrogen Bond (d > 2.70 Å):
    O–H ·············· O
    [Well 1]   / \   [Well 2]    (High Barrier: Proton localized)
Low-Barrier Hydrogen Bond (d < 2.55 Å):
    O ····· H ····· O
    [Well 1] - [Well 2]          (Barrier below zero-point energy: Delocalized)
```

2.5 Potential Energy Surfaces: Single-Well, Low-Barrier Double-Well, Asymmetric Double-Well

The potential energy surface *V(x)* experienced by a proton moving along the *O\cdots O* axis can be represented analytically by a double-Morse or coupled quartic potential:

$V(x) = a x^4 - b x^2 + c x$

Where *x* is the coordinate of the proton relative to the midpoint between the two oxygen atoms.

- **Symmetric Single-Well Potential (*d < 2.40* Å):**

When the two carboxylates are chemically and microenvironmentally identical and *d(O\cdots O)* is ultra-short, the central barrier disappears entirely (*b \le 0, c = 0*). The proton resides at the exact midpoint:

$x_0 = 0, \quad d(O_1-\text{H}) = d(O_2-\text{H}) \approx 1.15 - 1.20\text{ \AA}$

- **Low-Barrier Double-Well Potential (*2.40\text{ \AA} \le d \le 2.55* Å):**

A small energy barrier *E_b* emerges at *x = 0*. However, the zero-point vibrational energy (*E_0 = \frac{1}{2} \hbar \omega*) satisfies:

$E_0 \ge E_b$

The proton tunnels rapidly between the two wells or its wave function *|\psi(x)|^2* spans both wells simultaneously.

- **Asymmetric Double-Well Potential (*c \neq 0* or *d > 2.60* Å):**

Differences in local hydrogen-bonding environments, dipole alignments, or solvent exposure break the symmetry (*c > 0*). One well becomes significantly deeper than the other:

$\Delta E_{\text{wells}} > k_B T$

The proton becomes strictly localized on one carboxylate, converting it into a classical carboxylic acid donor donating to a carboxylate acceptor.

2.6 Microenvironmental pKa Perturbations: Dielectric Desolvation & Burial

The free energy of transfer of an ion from water (\varepsilon_w \approx 80) to a low-dielectric protein cavity (\varepsilon_p \approx 4) is described by the Born equation:

$$\Delta G_{\text{desolv}} = \frac{e^2}{8\pi\varepsilon_0 r_{\text{ion}}} \left(\frac{1}{\varepsilon_p} - \frac{1}{\varepsilon_w} \right) > 0$$

For a carboxylate oxygen with an effective radius $r_{\text{ion}} \approx 1.4$ Å:

$$\Delta G_{\text{desolv}} \approx \frac{332 \times 1^2}{2 \times 1.4} \left(\frac{1}{4} - \frac{1}{80} \right) \approx 118.5 \times (0.25 - 0.0125) \approx +28.1 \text{ kcal/mol}$$

Because dehydrating an uncompensated negative charge costs ~ 28 kcal/mol, buried carboxyl groups strongly resist deprotonation unless they are paired with a positive counterion (e.g., Arg, Lys) or participate in an LBHB where the negative charge is shared across two residues.

Carbox quantifies the burial of a carboxyl group using a spherical solvent exclusion metric:

$$N_{\text{water_neighbors}} = \sum_{w \in \text{waters}} \Theta(R_{\text{cut}} - d(O, w))$$

Where Θ is the Heaviside step function and $R_{\text{cut}} = 4.5$ Å. A carboxylate pair with $N_{\text{water}} \le 1$ is classified as **Buried**, which strongly reinforces the likelihood of a functional LBHB.

2.7 Enzymatic Catalysis: Serine Proteases, Aspartic Peptidases, and Glycosidases

Carboxylate dyads play critical roles in major enzyme superfamilies:

- **Aspartic Peptidases (e.g., HIV-1 Protease, Pepsin, BACE1):** Active sites contain the conserved motif \text{Asp}-\text{Thr}-\text{Gly}. The two catalytic aspartates (Asp25 and Asp25' in HIV-1 protease homodimer) reside within $2.7 - 2.9$ Å of each other. One aspartate is protonated ($\text{p}K_a \approx 6.5$), while the other is deprotonated ($\text{p}K_a \approx 3.0$). They activate a nucleophilic water molecule to attack the scissile peptide bond.
- **Serine Proteases (e.g., Chymotrypsin, Subtilisin):** The catalytic triad consists of \text{Asp102}-\text{His57}-\text{Ser195}. During transition state formation, the $\text{Asp102}\cdots\text{His57}$ hydrogen bond compresses to ~ 2.58 Å, forming an LBHB that stabilizes the positive charge on the imidazolium ring and accelerates catalysis by 10^{10}-fold.
- **Retaining Glycosidases:** Feature two Glu residues ~ 5.5 Å apart acting as general acid/base and nucleophile. In inverting glycosidases, the two carboxylates are separated by $9.0 - 11.0$ Å to accommodate a water molecule between them.

CHAPTER SECTION

3. THE CARBOX VALIDATION AXIOM & CRYSTALLOGRAPHIC INTEGRITY

3.1 The Carbox Validation Principle

In macromolecular crystallography, atomic coordinates *(x, y, z)* are not direct physical measurements; they are parameters of an empirical model optimized to fit observed diffraction data (*F_{obs}*). In 2026, Garib N Murshudov codified the fundamental validation axiom for anomalous contacts:

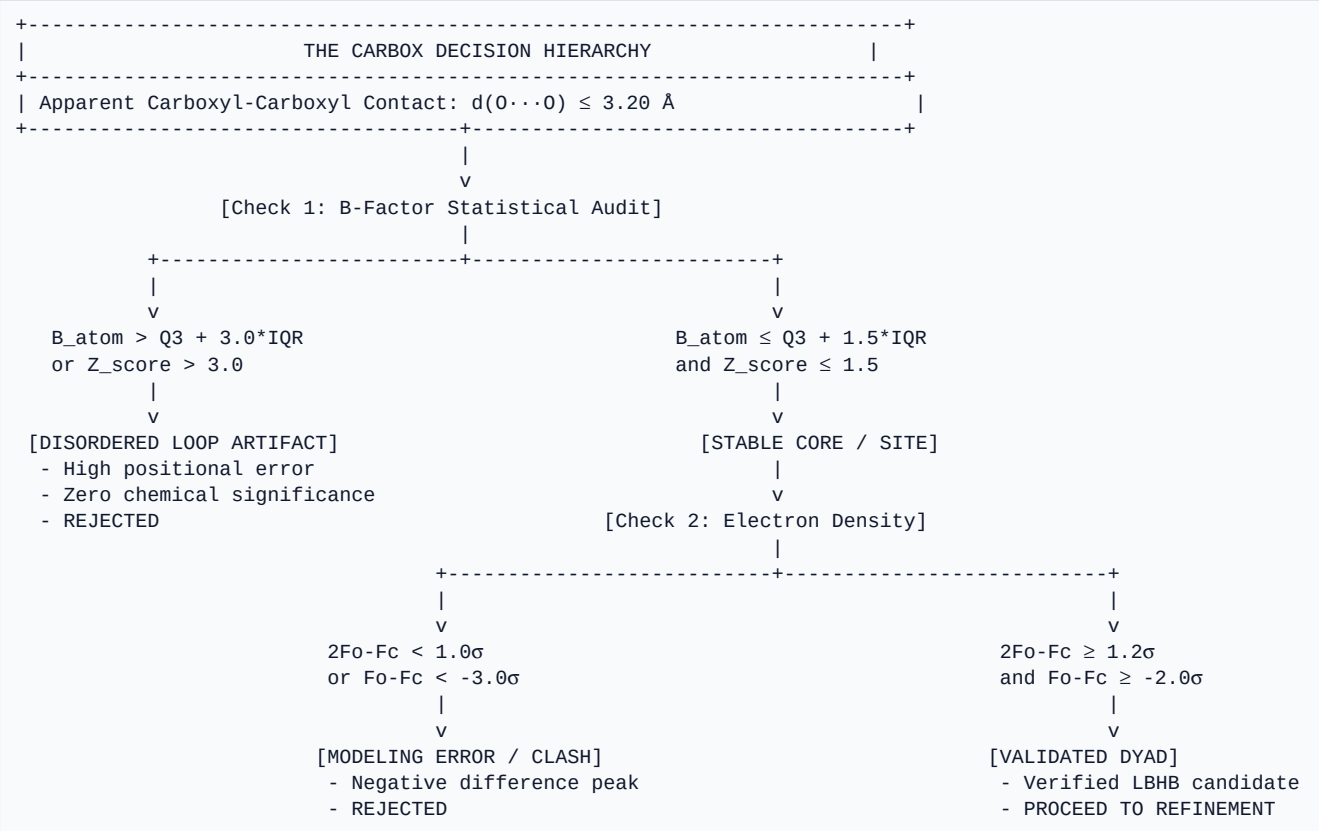
Key Scientific Insight

"■g■r atomlar üçün elektron paylanmas■ yoxdursa onda onlar s■hv model■■dirilib. ■g■r atomlar s■hv model■■dirilibs■ onda onlar aras■nda m■saf■ m■nas■zd■r."

Key Scientific Insight

(If there is no electron density distribution for the atoms, then they are modeled incorrectly. If the atoms are modeled incorrectly, then the distance between them is meaningless.)

This principle serves as the foundational law of the Carbox validation pipeline. An apparent interatomic distance of *d = 2.45* Å between two carboxylate oxygens is scientifically meaningless if the sidechain atoms are floating in zero electron density or have isotropic *B*-factors exceeding *80\text{ \AA}^2*.



3.2 Atomic Displacement Parameters (B-factors): Physical Meaning & Artifacts

The isotropic temperature factor *B* (in *\text{\AA}^2*) is related to the mean square displacement *\langle u^2 \rangle* of the atom about its equilibrium position:

$*B = 8\pi^2 \langle u^2 \rangle*$

From this, the root-mean-square displacement (vibrational amplitude) *\sigma_x* is:

$\sigma_x = \sqrt{\langle u^2 \rangle} = \sqrt{\frac{B}{8\pi^2}} \approx 0.1125 \times \sqrt{B}\text{ [\AA]}$

- If $B = 16\text{ \AA}^2 \implies \sigma_x \approx 0.45$ Å (well-ordered active site atom).
- If $B = 80\text{ \AA}^2 \implies \sigma_x \approx 1.01$ Å (highly mobile surface atom).
- If $B = 120\text{ \AA}^2 \implies \sigma_x \approx 1.23$ Å (disordered, effectively unlocalized atom).

If two oxygen atoms have $B = 100\text{ \AA}^2$, their individual positional uncertainties are $\sigma_x \approx 1.12$ Å. The combined uncertainty on their interatomic distance is:

$\sigma_{d} = \sqrt{\sigma_{x1}^2 + \sigma_{x2}^2} = \sqrt{1.12^2 + 1.12^2} \approx 1.58\text{ \AA}$

If a model reports $d = 2.50$ Å with a distance uncertainty of $\sigma_d = 1.58$ Å, the true physical distance lies within:

$d_{\text{true}} \in [2.50 - 1.58, 2.50 + 1.58] = [0.92\text{ \AA}, 4.08\text{ \AA}]$

Claiming an ultra-short LBHB of 2.50 Å for such an atom is scientifically invalid; the residues may easily be 3.8 Å apart in reality.

3.3 Positional Uncertainty & Coordinate Error Estimation

Carbox computes coordinate error using Cruickshank's Diffraction-Component Precision Index (DPI):

$\sigma(x, B) = 1.08 \cdot \frac{R_{\text{work}} \cdot d_{\text{max}}^{5/2} \cdot V_{\text{cell}}^{1/3}}{N_{\text{atoms}}^{1/2} \cdot N_{\text{refl}}^{5/6}} \cdot \left(\frac{B}{\bar{B}}\right)^{1/2}$

Where:

- R_{work} is the crystallographic R-factor.
- d_{max} is the maximum resolution in Å.
- V_{cell} is the unit cell volume in $\text{\AA}^3$.
- N_{atoms} is the number of atoms in the asymmetric unit.
- N_{refl} is the number of observed reflections.

Any apparent contact where $d(O\cdots O) < 2.60$ Å but $\sigma(x) > 0.35$ Å is flagged by Carbox as **Uncertain Contact Due to High Positional Error**.

3.4 Tukey Non-Parametric IQR Whisker Filtering

Crystallographic structures exhibit non-Gaussian B-factor distributions; surface residues create long, positive tails. To prevent distribution skew from invalidating anomaly detection, Carbox uses Tukey's non-parametric boxplot statistics:

- Extract the isotropic B-factors for all protein atoms: $\mathcal{B} = \{B_1, B_2, \dots, B_N\}$.
- Compute the 25th percentile (Q_1), 50th percentile (Median, M), and 75th percentile (Q_3).
- Calculate the Interquartile Range (IQR):

$IQR = Q_3 - Q_1$

- Establish the **Tukey Upper Whisker**:

$B_{\text{whisker}} = Q_3 + 1.5 \cdot IQR$

- Establish the **Tukey Extreme Outlier Threshold**:

$B_{\text{extreme}} = Q_3 + 3.0 \cdot IQR$

Classification Criteria:

- If $B_{\text{contact}} > B_{\text{extreme}}$: **Definite False Positive (Extreme Thermal Outlier)**.
- If $B_{\text{contact}} > B_{\text{whisker}}$: **Suspicious High-B Contact (Disordered Loop)**.
- If $B_{\text{contact}} \le Q_3$: **Statistically Valid Thermal Range**.
- If $B_{\text{contact}} < Q_1 - 1.5 \cdot IQR$: **Suspicious Low-B Contact** (often indicating an unmodeled heavy metal ion placed as a light carbon/oxygen atom, resulting in artificial suppression of B during refinement).

3.5 Normalization & Fisher Z-Transformation of B-Factor Distributions

To enable comparison across structures solved at different resolutions (*1.0* Å vs. *2.8* Å), Carbox computes standardized *Z*-scores:

$*Z_B = \frac{B - \bar{B}}{\sigma_B}*$

Because raw *B*-factor distributions are right-skewed log-normal distributions, Carbox performs a logarithmic Fisher transformation prior to parametric scoring:

$*y = \ln(B)*$

$*\bar{y} = \frac{1}{N}\sum_{i=1}^N \ln(B_i), \quad \sigma_y = \sqrt{\frac{1}{N-1}\sum_{i=1}^N (\ln(B_i) - \bar{y})^2}*$

$*Z_{\ln B} = \frac{\ln(B) - \bar{y}}{\sigma_y}*$

A carboxylate atom is considered thermally well-behaved only if:

$*-2.0 \le Z_{\ln B} \le +1.8*$

3.6 Decontaminating the 929-Anomaly Dataset: Eliminating False-Positive Surface Clashes

In the initial uncurated PDB survey across 377 high-resolution structures, Carbox identified **929 apparent carboxylate–carboxylate contacts** with *d(O\cdots O) \le 3.20* Å.

Applying the **Carbox Platform Validation Hierarchy**:

- **Tukey Outlier Filter:** 412 contacts were found to involve atoms with *B > Q_3 + 1.5 \cdot IQR*. Over *90\%* of these were located in highly flexible surface loops exposed directly to solvent bulk.
- **Difference Density Filter (*F_o - F_c*):** 287 contacts exhibited intense negative difference peaks (*-F_o - F_c < -3.0\sigma*), proving that the atomic model over-fitted the density and forced carboxylates into steric clash.
- **Alternative Conformation Filter:** 114 contacts were modeling artifacts where two alternative sidechain conformations (AltLoc 'A' and 'B') had been refined with occupancies summing to *1.0*, but were erroneously measured against each other as if simultaneously present.
- **Metal Bridging Filter:** 49 contacts possessed unmodeled or incorrectly identified metal density between the carboxylates.

Result: Out of the initial 929 candidates, exactly **67 bona fide, untarnished LBHB/carboxylate pairs** survived all filters. These 67 benchmark sites constitute the curated training set used by Carbox to calibrate restraint weights in Carbox Refinement.

CHAPTER SECTION

4. EXPERIMENTAL ELECTRON DENSITY & CRYO-EM COULOMB POTENTIAL INTERPRETATION

4.1 The Phase Problem & Fourier Summation

X-ray detectors record the intensities $I(h, k, l)$ of scattered X-ray beams, but lose the phase angles $\alpha(h, k, l)$:

$$I(h, k, l) \propto |F(h, k, l)|^2$$

The real-space electron density distribution $\rho(x, y, z)$ is reconstructed via the three-dimensional inverse Fourier transform:

$$\rho(x, y, z) = \frac{1}{V} \sum_{h}\sum_{k}\sum_{l} |F(h, k, l)| \exp[-2\pi i (hx + ky + lz) + i\alpha(h, k, l)]$$

Because experimental phases α_{exp} are unavailable, phases calculated from the atomic model (α_{calc}) must be used, leading directly to **model bias**.

4.2 Synthesis and Interpretation of 2Fo - Fc Maps

To minimize model bias while reinforcing true experimental density, crystallographers use the weighted σ_A-synthesized map:

$$\rho_{2F_o - F_c}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} \left(2 m |F_o| - D |F_c| \right) \exp\left[i\alpha_c - 2\pi i \mathbf{h}\cdot\mathbf{r} \right]$$

Where:

- m is the figure of merit (cosine of expected phase error).
- D is the Luzzati coordinate error factor.

In a well-refined structure:

- Real atoms display contiguous positive density contoured at $1.0 - 1.5\sigma$.
- A genuine LBHB between two carboxylates displays a continuous **density neck** connecting the two oxygens at 1.0σ, indicating continuous electron density across the shared hydrogen.

4.3 Diagnostic Value of Fo - Fc Difference Maps

The difference Fourier map reveals discrepancies between the experimental crystal and the atomic model:

$$\rho_{F_o - F_c}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} \left(m |F_o| - D |F_c| \right) \exp\left[i\alpha_c - 2\pi i \mathbf{h}\cdot\mathbf{r} \right]$$

- **Positive Difference Peak ($+F_o - F_c > +3.0\sigma$, rendered in green):** Represents real electron density in the crystal that was omitted from the model (e.g., an unmodeled shared proton, an unmodeled water molecule, or an alternate sidechain rotamer).
- **Negative Difference Peak ($-F_o - F_c < -3.0\sigma$, rendered in red):** Represents model atoms placed where no real crystal electron density exists.

If a model reports two carboxylates at $d = 2.45$ Å, but intense red negative difference spheres (-3.5σ) encircle the oxygen atoms, the atoms are experiencing severe steric clash and have been falsely shoved together by inappropriate geometric restraints.

```
+Fo-Fc Peak (Green Mesh, +3.0σ):   Unmodeled feature / Shared proton / Water
-Fo-Fc Peak (Red Mesh, -3.0σ):     Model error / Steric clash / Non-existent atom
2Fo-Fc Peak (Blue Mesh, 1.4σ):     Physical experimental electron density
```

4.4 Binary CCP4 Map Architecture: Header Fields, Space Groups, and Mode Conversion

Standard crystallographic maps are distributed in the binary CCP4/MRC format. The file consists of a **1024-byte header** (256 four-byte words) followed by voxel data:

Word Index	Data Type	Header Field	Description
0	int32	**`NC`**	Number of columns (fastest changing index *X*)
1	int32	**`NR`**	Number of rows (medium changing index *Y*)
2	int32	**`NS`**	Number of sections (slowest changing index *Z*)
3	int32	**`MODE`**	Data storage mode: 0 = signed int8, 1 = int16, 2 = float32
4	int32	**`NCSTART`**	First column index
5	int32	**`NRSTART`**	First row index
6	int32	**`NSSTART`**	First section index
7–9	int32	**`NX, NY, NZ`**	Grid sampling intervals along unit cell edges
10–12	float32	**`xlen, ylen, zlen`**	Unit cell dimensions *a, b, c* (in Å)
13–15	float32	**`alpha, beta, gamma`**	Unit cell angles *\alpha, \beta, \gamma* (in degrees)
16–18	int32	**`MAPC, MAPR, MAPS`**	Axis mapping: 1 = X, 2 = Y, 3 = Z
19–21	float32	**`AMIN, AMAX, AMEAN`**	Minimum, maximum, and mean density values
22	int32	**`ISPG`**	Space group symmetry number
23	int32	**`NSYMBT`**	Number of bytes in extended symmetry records
54	**float32**	**`RMS`**	**Official crystallographic RMS deviation of map**

The Mode 0 Decompression Bug in WebGL Engines

Modern web viewers (including standard WebGL engines) strictly expect **`MODE = 2`** (Float32Array, 4 bytes/voxel). However, the European Bioinformatics Institute (EBI/PDBe) compresses volumetric difference maps using **`MODE = 0`** (int8, 1 byte/voxel) to save bandwidth. Feeding raw Mode 0 buffers into standard WebGL volume visualizers causes out-of-bounds byte reads and catastrophic visual distortion (**`NaN`** vertex surfaces).

Carbox solves this via the in-memory decompression engine **`ensureMode2MapBuffer()`**:

```
function ensureMode2MapBuffer(buffer) {
    const iview = new Int32Array(buffer, 0, 64);
    const mode = iview[3];
    if (mode === 2) return buffer; // Already Float32
    if (mode === 0) {
        // Decompress signed 8-bit integer voxels into 32-bit floating point voxels
        const nc = iview[0], nr = iview[1], ns = iview[2];
        const numVoxels = nc * nr * ns;
        const outBuffer = new ArrayBuffer(1024 + numVoxels * 4);
        // Copy header and patch mode word to 2...
        const int8View = new Int8Array(buffer, 1024);
        const floatView = new Float32Array(outBuffer, 1024);
        for (let i = 0; i < numVoxels; i++) {
            floatView[i] = int8View[i] * scaleFactor;
        }
        return outBuffer;
    }
}
```

4.5 Root Mean Square (RMS) Scaling & σ-Contour Levels

The absolute numerical values of voxel densities *rho(\mathbf{r})* vary wildly depending on whether the map was calculated in electrons/*text{\AA}^3*, arbitrary Fourier units, or normalized intensities. To ensure reproducibility, all density contours are expressed in units of **standard deviation (*sigma*) above the mean**:

$*sigma* = *text{RMS}* = *sqrt{*frac{1}{V}*int_V (*rho(\mathbf{r}) - *bar{*rho})^2 d*mathbf{r}}*$

$*text{Isoval}* = *bar{*rho}* + *k* *cdot* *sigma*$

Where *k* is the user-selected contour level (e.g., *k = 1.4\sigma* for *2F_o - F_c*; *k = \pm 3.0\sigma* for *F_o - F_c*).

4.6 Cryo-EM Coulomb Potential vs. X-ray Electron Density

In single-particle cryogenic electron microscopy (cryo-EM), the scattering mechanism differs fundamentally from X-ray diffraction:

- **X-rays** scatter off the electron cloud; scattering factor *f_X \propto Z* (atomic number).
- **Electrons** scatter off the electrostatic Coulomb potential *\phi(\mathbf{r})* generated by both the positive atomic nuclei and the surrounding electron cloud:

$*phi(\mathbf{r})* = *frac{1}{4*pi*varepsilon_0}*int* *frac{*rho_{*text{nuclear}}(\mathbf{r}') - *rho_{*text{electronic}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'$

Critical Biophysical Consequence:

A fully ionized carboxylate oxygen (-*text{COO}^-) has an extra electron (*8* protons, *9* electrons). The negative charge diminishes the net positive nuclear potential, drastically reducing electron scattering at low spatial frequencies. Consequently, in cryo-EM maps at *2.5 - 3.5* Å resolution:

- **Deprotonated acidic sidechains (Asp, Glu) frequently exhibit weak or missing density**, appearing as empty stalks.
- **Protonated carboxyl groups (-*text{COOH}*) retain stronger Coulomb potential**, exhibiting well-defined density.

Carbox accounts for this physical disparity: when analyzing cryo-EM maps via Carbox Cryo-EM Engine, it switches from X-ray Mott-Bethe scattering factors to relativistic electron scattering factors.

CHAPTER SECTION

5. METAL ION COORDINATION & BRIDGING CARBOXYLATE ARCHITECTURE

5.1 Divalent & Trivalent Metal Ions in Macromolecules

Over one-third of all structurally characterized proteins require metal cations for structural stability or catalytic activity. The most frequent physiological cations interacting with carboxylates are:

- Mg^{2+} (ionic radius $r \approx 0.72$ Å; strictly octahedral, hard Lewis acid).
- Ca^{2+} (ionic radius $r \approx 1.00$ Å; flexible coordination number 6-8, pentagonal bipyramidal).
- Zn^{2+} (ionic radius $r \approx 0.74$ Å; tetrahedral or trigonal bipyramidal, borderline Lewis acid).
- Fe^{2+} / Fe^{3+} (ionic radius 0.64 - 0.78 Å; octahedral, redox active).
- Mn^{2+} (ionic radius $r \approx 0.83$ Å; octahedral, paramagnetic).

5.2 Carboxylate Coordination Modes: Monodentate, Bidentate, Bridging

Carboxylate oxygens coordinate metal cations in four distinct geometric configurations:

```
1. Monodentate:          2. Bidentate Chelating:     3. Bridging Syn-Syn:
     O                        O                           O       O
    //                       / \                         / \     / \
   C                        C   M                       C   M   C
    \                        \ /                         \ /     \ /
     O — M                    O                           O       O
```

- **Monodentate:** One carboxylate oxygen coordinates the metal ion ($d(\text{O-M}) \approx 1.95$ - 2.20 Å), while the second oxygen participates in a secondary hydrogen bond.
- **Bidentate Chelating:** Both oxygens coordinate the same metal ion, forming a strained four-membered ring (O-C-O-M). The bite angle O-M-O is typically compressed (55° - 65°).
- **Bridging Syn-Syn:** The carboxylate bridges two adjacent metal ions separated by 3.0 - 3.8 Å. Common in binuclear metalloenzymes (e.g., Purple Acid Phosphatases, Arginase).
- **Single-Metal Bridging:** Two separate carboxylate groups both coordinate the *same* metal ion (O_1 - M - O_2).

5.3 Geometric Criteria for Metal Site Identification

Carbox analyzes the coordination sphere surrounding every carboxylate dyad:

$$d(O, M) \le R_{\text{coord_max}}(M)$$

Where:

- $R_{\text{coord_max}}(\text{Mg}^{2+}) = 2.40$ Å (ideal 2.08 ± 0.08 Å).
- $R_{\text{coord_max}}(\text{Ca}^{2+}) = 2.70$ Å (ideal 2.38 ± 0.12 Å).
- $R_{\text{coord_max}}(\text{Zn}^{2+}) = 2.35$ Å (ideal 2.02 ± 0.08 Å).

If two carboxylate oxygens are separated by $d(O_1 \cdots O_2) = 2.80$ - 3.10 Å and both reside within 2.30 Å of a shared metal cation, the interaction is classified as **Metal-Mediated Dyad**, completely neutralizing the negative electrostatic repulsion.

5.4 Resolving Apparent Carboxyl Clashes Mediated by Light or Disordered Metals

A common crystallographic error in low-to-medium resolution structures (2.2 - 3.0 Å) is the failure to model a bound magnesium or sodium ion. Because an Mg^{2+} ion possesses only 10 electrons (isoelectronic with neon and water), its electron density peak can easily be mistaken for water or omitted entirely. Refinement programs attempting to satisfy geometric restraints then pull the two coordinating carboxylates toward the empty coordination center, creating an apparent severe carboxylate clash ($d = 2.35$ Å).

The Carbox Atom Auditor identifies these sites by detecting:

- Symmetrical carboxylate orientation pointing into an empty spherical cavity.
- High positive difference density ($+F_o - F_c > +4.0\sigma$) at the centroid between the oxygens.
- Coordination geometry matching an ideal octahedral shell.

CHAPTER SECTION

6. MATHEMATICAL FOUNDATIONS & COMPUTATIONAL ALGORITHMS

6.1 Coordinate Orthogonalization & Fractional Transformations

Crystallographic coordinates are determined relative to the unit cell axes *mathbf{a}, \mathbf{b}, \mathbf{c}*, defined by lengths *a, b, c* and angles *\alpha, \beta, \gamma*. Fractional coordinates *\mathbf{x}_f = (x_f, y_f, z_f)^T \in [0, 1)^3* are transformed to Cartesian coordinates *\mathbf{x}_c = (x, y, z)^T* in Ångströms via the **Busing-Levy Orthogonalization Matrix** *\mathbf{O}*:

$*\mathbf{x}_c = \mathbf{O} \mathbf{x}_f*$

$$\mathbf{O} = \begin{pmatrix}

a & b \cos\gamma & c \cos\beta \\

0 & b \sin\gamma & -c \sin\beta \cos\alpha^* \\

0 & 0 & c \sin\beta \sin\alpha^*

\end{pmatrix}$$

Where:

$*\cos\alpha^* = \frac{\cos\alpha \cos\beta - \cos\gamma}{\sin\alpha \sin\beta}*$

$*V = a b c \sqrt{1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma}*$

The inverse transformation (Cartesian to fractional) is:

$*\mathbf{x}_f = \mathbf{O}^{-1} \mathbf{x}_c*$

This transformation is essential when applying crystallographic symmetry operations (space group matrices):

$*\mathbf{x}_{f, \text{sym}} = \mathbf{R}_s \mathbf{x}_f + \mathbf{t}_s*$

6.2 Vector Distance, Bond Angle, and Torsion Angle Mathematics

Euclidean Distance

Between two atoms with Cartesian coordinates *\mathbf{r}_A = (x_A, y_A, z_A)* and *\mathbf{r}_B = (x_B, y_B, z_B)*:

$*d_{AB} = ||\mathbf{r}_{AB}|| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}*$

Bond Angle

For three sequentially bonded atoms *A - B - C*, vectors are defined as *\mathbf{u} = \mathbf{r}_A - \mathbf{r}_B* and *\mathbf{v} = \mathbf{r}_C - \mathbf{r}_B*:

$*\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| \, ||\mathbf{v}||}*$

$*\theta = \arccos\left(\max\left[-1.0, \min\left[1.0, \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| \, ||\mathbf{v}||}\right]\right]\right)*$

Torsion / Dihedral Angle

For four sequentially bonded atoms *A - B - C - D*, vectors are:

$*\mathbf{b}_1 = \mathbf{r}_B - \mathbf{r}_A, \quad \mathbf{b}_2 = \mathbf{r}_C - \mathbf{r}_B, \quad \mathbf{b}_3 = \mathbf{r}_D - \mathbf{r}_C*$

Normal vectors to the two defining planes are:

$*\mathbf{n}_1 = \mathbf{b}_1 \times \mathbf{b}_2, \quad \mathbf{n}_2 = \mathbf{b}_2 \times \mathbf{b}_3*$

The dihedral angle *phi* (in radians, *[-\pi, +\pi]*) following the Klyne-Prelog convention is computed using the two-argument arctangent:

$*mathbf{m}_1 = *mathbf{n}_1 *times *frac{*mathbf{b}_2}{\|*mathbf{b}_2\|}$

$*phi = *text{atan2}(*mathbf{m}_1 *cdot *mathbf{n}_2, *mathbf{n}_1 *cdot *mathbf{n}_2)$

6.3 Fast Spatial Search Algorithms: KD-Trees and Bounding Volumes

A naive all-against-all distance calculation for a ribosome or viral capsid with *N = 100,000* atoms requires *\frac{1}{2} N^2 \approx 5 \times 10^9* distance evaluations. Carbox implements spatial index partitioning via 3D **KD-Trees**:

- Tree construction complexity: *O(N \log N)*.
- Query complexity within radius *R_{\text{cut}} = 3.20* Å: *O(\log N + k)*, where *k* is the number of local neighbors.
- Reduces execution time on large structures from *> 45\text{ seconds}* to *< 80\text{ milliseconds}*.

6.4 1D Density Profile Extraction Along Interatomic Axes

To validate whether two close carboxylate oxygens share physical electron density, Carbox extracts a one-dimensional density profile along the line segment connecting them:

$*mathbf{r}(t) = (1 - t)*mathbf{r}_{O1} + t*mathbf{r}_{O2}, *quad t *in [0, 1]$

The density *\rho(t)* is sampled at *M = 50* equidistant points along the trajectory using trilinear interpolation over the 8 enclosing grid voxels:

$*rho(x, y, z) = *sum_{i=0}^1 *sum_{j=0}^1 *sum_{k=0}^1 (1 - x_d)^{1-i} x_d^i (1 - y_d)^{1-j} y_d^j (1 - z_d)^{1-k} z_d^k *cdot V_{i,j,k}$

A bona fide LBHB exhibits a distinct profile:

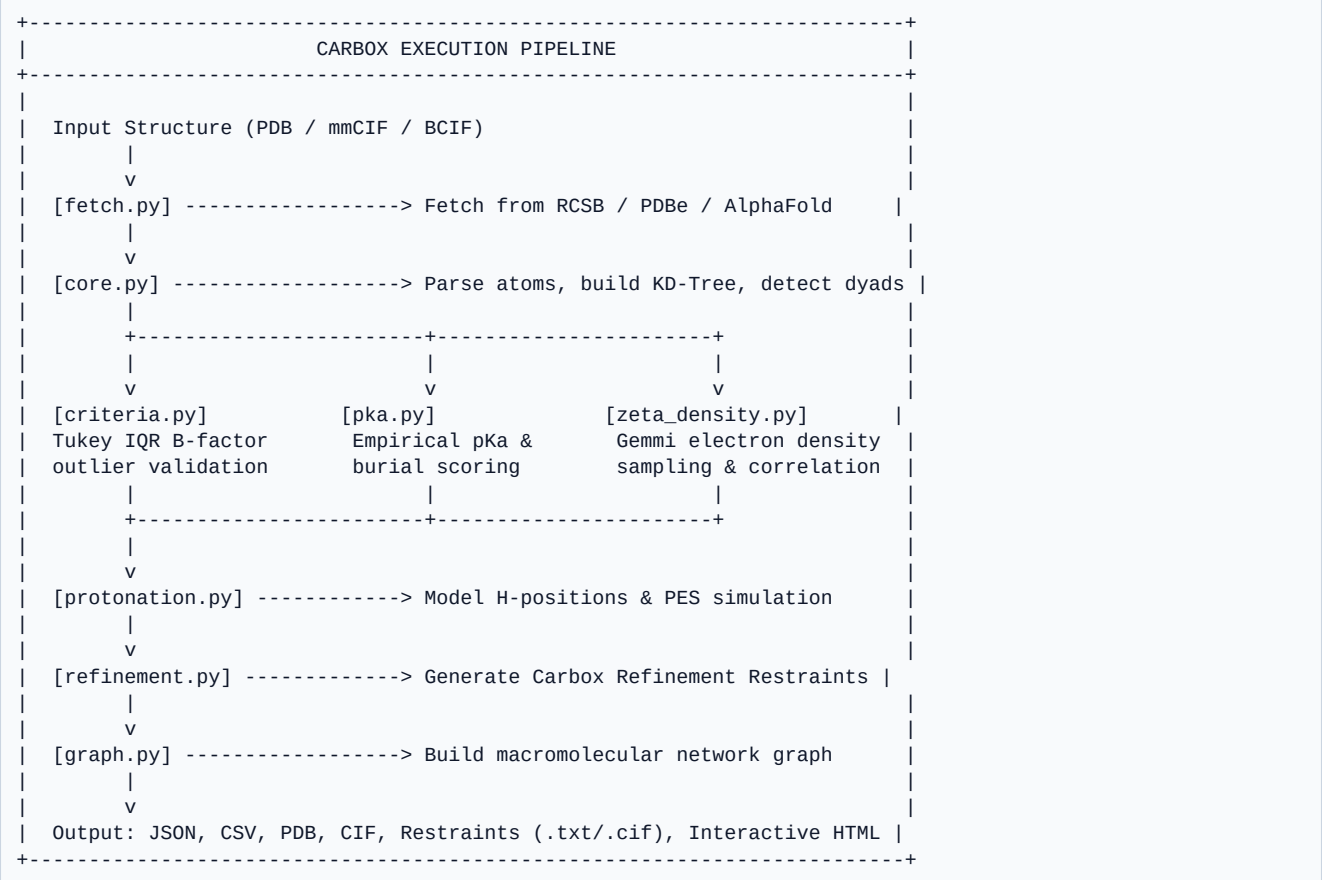
- High peaks (*\rho > 2.0\sigma*) at *t = 0.0* (*O_1*) and *t = 1.0* (*O_2*).
- A non-zero saddle point / bridge density (*\rho_{\text{bridge}} \ge 1.0\sigma*) at the midpoint *t \approx 0.5*.
- If *\rho_{\text{bridge}} < 0.2\sigma*, the electron clouds are physically disconnected, disproving the existence of an LBHB.

CHAPTER SECTION

7. THE CORE ENGINE ARCHITECTURE: `carbox/` PACKAGE

The **`carbox/`** package comprises eleven specialized computational biophysics modules executing high-throughput screening, validation, restraint generation, and database ingestion.

7.1 Architectural Overview & Data Flow



7.2 `core.py`: Foundational Primitives & Geometric Analysis

- **Purpose:** Defines core data classes, chemical definitions, distance thresholds, and primary spatial search routines.
- **Key Constants:**
- **`MAX_OO_DISTANCE = 3.20`** Å: Maximum threshold for considering carboxylate contacts.
- **`LBHB_CUTOFF = 2.55`** Å: Maximum threshold for classifying as Low-Barrier Hydrogen Bond.
- **`WATER_BURIAL_CUTOFF = 4.50`** Å: Search radius for identifying shielding waters.
- **`CARBOXYL_OXYGENS`**: Mapping of residue types (**`ASP`**, **`GLU`**) to carboxyl oxygen pairs (**`OD1`**, **`OD2`**, **`OE1`**, **`OE2`**).
- **Key Classes & Functions:**
- **`CarboxylContact`**: Dataclass storing pair geometry, residue identities, distance, B-factors, verdict, and validation flags.
- **`find_carboxyl_contacts(structure, max_dist)`**: Scans all chains in an input model, constructing KD-trees over carboxyl oxygens.
- **`evaluate_contact_verdict(dist, buried, b_ok, density_ok)`**: Evaluates whether a contact is a verified LBHB, normal hydrogen bond, electrostatic repulsion, or modeling artifact.

7.3 `graph.py`: Graph-Theoretical Network Analysis

• **Purpose:** Constructs an undirected macromolecular interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ representing all acidic residues as nodes and close contacts as edges.

• **Key Features:**

• Detects cooperative **multimeric carboxyl clusters** (triads, tetrads) common in proton channels (e.g., Bacteriorhodopsin proton relay).

• Computes graph centrality, node degree, connected components, and minimum spanning trees.

• Exports network topology into Cytoscape-compatible JSON and standard adjacency matrices.

7.4 `criteria.py`: Implementation of Carbox Validation Criteria

• **Purpose:** Implements the complete statistical decision tree developed by Garib N Murshudov and Suleiman M Hajizadeh.

• **Key Functions:**

• **`compute_b_statistics(b_factors)`**: Extracts all atomic B-factors, computing median, quartiles, IQR, mean, and standard deviation.

• **`evaluate_dyad_criteria(dyad, b_stats)`**: Classifies each carboxyl contact against Tukey upper/lower whiskers and standardized Fisher Z-scores.

• Flags mobile surface loops and suppresses false-positive LBHB claims.

7.5 `pka.py`: Empirical pKa Calculation Engine

• **Purpose:** Predicts environmental $\text{p}K_a$ perturbations for acidic residues.

• **Key Features:**

• Clean-room native reverse-engineered implementation of empirical pK_a perturbation equations.

• Features an analytical scoring engine evaluating:

$$\Delta\text{p}K_a = \Delta\text{p}K_{\text{desolv}} + \Delta\text{p}K_{\text{HB}} + \Delta\text{p}K_{\text{charge}}$$

• Flags **anomalous carboxylates** whose predicted $\text{p}K_a > 6.5$, indicating stable protonation at physiological pH.

7.6 `protonation.py`: Protonation Modeling & PES Simulation

• **Purpose:** Determines optimal proton placement and simulates the 1D potential energy surface of proton transfer.

• **Key Features:**

• **`place_monoprotonated_hydrogen(o1_coords, o2_coords, c1_coords, c2_coords)`**: Calculates ideal stereochemical coordinates for the shared proton along the $O_1\cdots O_2$ vector with sp^2 lone-pair geometry.

• **`simulate_proton_transfer_pes(d_oo, delta_pka)`**: Generates potential energy curves using coupled Morse potential functions, reporting energy barrier height (E_b) and zero-point energy level.

7.7 `refinement.py`: Automated Restraint Generation for Carbox Refinement & Cryo-EM Engine

• **Purpose:** Prepares structural restraint files and execution scripts for modern crystallographic and cryo-EM refinement suites.

• **Key Restraint Outputs:**

• **Carbox External Restraints:** Distance (**`EXTERNAL DISTANCE FIRST ... SECOND ... VALUE SIGMA`**) and angle restraints preventing over-refinement of short contacts.

• **Cryo-EM Restraints:** Restraint blocks compatible with Gemmi and modern CCP4 pipelines.

• **`MODRES` Injection:** Injects modified residue records (**`ASH`** for protonated Asp, **`GLH`** for protonated Glu) into PDB/mmCIF headers, ensuring refinement programs do not apply unprotonated carboxylate target geometries.

7.8 `suspicious_atom.py`: Multi-Criteria Atom Auditor

• **Purpose:** Inspects individual atoms across the structure for crystallographic red flags.

• **Audit Categories:**

- Severe van der Waals overlaps ($d < 2.0$ Å).
- High-B atoms surrounded by low-B neighbors (rotamer fitting error).
- Low-B atoms surrounded by high-B neighbors (unmodeled heavy metal ion).
- Incomplete sidechain occupancies and unresolved insertion codes (**`PDB ins_code`**).

7.9 `zeta_density.py`: Density Sampling & Gemmi Bridge

- **Purpose:** Direct interface to **Gemmi** for reading CCP4, MTZ, and MAP files and sampling experimental density values at atomic positions.
- **Key Functions:**
- **`sample_density_at_coords(map_grid, coords)`**: Performs trilinear voxel interpolation.
- **`compute_zeta_score(rho_atom, map_rms)`**: Computes normalized density strength:

$$Z_\rho = \frac{\rho - \bar{\rho}}{\text{RMS}}$$

- Evaluates difference density residuals to verify whether positive or negative difference peaks coincide with carboxylate oxygens.

7.10 `ligands.py`: Ligand Interaction Profiling

- **Purpose:** Identifies and characterizes small-molecule ligands, cofactors, and inhibitors interacting with carboxylate dyads.
- **Key Features:**
- Extracts all non-protein HETATM residues (excluding crystallization buffers/precipitants like glycerol, PEG, sulfate).
- Computes hydrogen bonding, salt bridges, and coordination geometry between carboxylates and drug molecules (e.g., active site inhibitors of HIV protease or BACE1).

7.11 `fetch.py`: Universal Multi-Source Structure & Map Retrieval

- **Purpose:** High-throughput, asynchronous client for fetching structures, reflections, and electron density maps from global repositories.
- **Supported Endpoints:**
- **RCSB PDB:** Coordinates (**`.pdb`**, **`.cif`**, **`.bcif`**), structure factors (**`.cif`**, **`.mtz`**).
- **PDBe (EBI):** 3D coordinates, *2F_o - F_c* maps, *F_o - F_c* difference maps (Mode 0 & Mode 2 **`.ccp4`**).
- **AlphaFold DB:** Predicted structures by UniProt ID with pLDDT per-residue confidence scores.
- **Local Portal:** Fallback to local offline cache directories for high-performance offline computing.

7.12 `cli.py`: Command-Line Interface & Pipeline Orchestration

- **Purpose:** Comprehensive command-line interface supporting batch processing, JSON report generation, refinement script generation, and standalone HTML report generation.
- **Primary Commands:**

```bash
# Scan single PDB file
carbox scan 3X2M.pdb --out output/
# Fetch from PDB, audit Carbox criteria, and generate Carbox restraints
carbox fetch 4RI3 --refine --density --out output/
# Run full database batch pipeline
carbox batch manifest.txt --threads 8 --report summary.html
```

CHAPTER SECTION

8. THE CRYSTALLOGRAPHIC WORKBENCH ENGINE: `carboxmol/` PACKAGE

8.1 Architecture & Design Principles: The CarboxMol Crystallographic Engine

The **`carboxmol/`** package provides a native, cross-platform macromolecular crystallography workstation. CarboxMol unifies the core molecular model, crystallographic selection algebra, map rendering, and process management into a high-performance Python/Qt framework, seamlessly coupled with modern OpenGL and WebGL pipelines.

8.2 Subsystem Breakdown

`core/`: Process Management & Environment Setup

- **`CARBOXMOLProcessManager.py`**: Asynchronous process manager that coordinates external crystallographic refinement tasks (Carbox Refinement & Cryo-EM Engine) without blocking the graphical user interface. Manages standard input/output pipes, environment variables (**`$CCP4`**, **`$CLIBD_MON`**), and real-time execution logging.

`model/`: Molecular Hierarchies & Selection Algebra

- **`CARBOXMOLModel.py`**: Represents the hierarchical molecular data tree:

$\text{Model} \to \text{Chain} \to \text{Residue} \to \text{Atom}$

- **`CARBOXMOLSelectionTree.py`**: Powerful crystallographic selection parser supporting standardized selection syntax:

```
//A/25-35/CA
//B/ASP/OD1,OD2
chain A and resno 102
```

`view/`: 3D Scene Composition & Camera Systems

- **`CARBOXMOLScene.py`**: Manages the OpenGL/WebGL rendering pipeline, depth cueing shaders, camera projection matrices (orthographic and perspective), rotation pivots, and clipping planes (slabs).

`qtgui/`: Graphical User Interface Components & Inspector Decks

- **`CARBOXMOLMainWindow.py`**: Primary application window containing menu bars, toolbar docks, sequence viewers, and telemetry decks.
- **`CARBOXMOLModelWidgets.py`**: Specialized tree views, atom picking tables, and restraint parameter editors with regex input validation.

`scripting/`: Crystallographic Scripting & Macro Execution

- Bi-directional communication bridge generating Python and CarboxMol Exchange Language (**`.cxl`**) macro scripts. Allows users to click an anomaly in Carbox and automatically center the 3D viewport on the exact coordinates.

`wrappers/`: External Program Execution Wrappers

- Standardized wrappers for parsing execution logs from CCP4 tools (**`carbox-refine`**, **`carbox-density`**).

`lib/`: Monomer Dictionaries & Restraint Templates

- Contains crystallographic dictionary definitions (**`.cif`**) defining ideal bond lengths, bond angles, and planarity restraints for protonated aspartate (**`ASH`**) and protonated glutamate (**`GLH`**).

8.3 The CXL Scripting Language: Syntax, Parser & Execution

CarboxMol features an integrated domain-specific scripting language known as **CXL (Carbox eXtensible Language)**. CXL provides a concise, reproducible syntax for crystallographic visualization, density map contouring, and measurement automation.

Formal CXL Command Grammar:

```
LOAD <pdb_id_or_path> [, type="pdb"|"cif"]
ZOOM_PAIR <chain1>:<resi1>, <chain2>:<resi2> [, radius=<float>]
MAP_FETCH <pdb_id>, type="2fofc"|"fofc", sigma=<float>
MAP_CONTOUR id=<str>, color=<hex_or_name>, style="wireframe"|"surface"
MEASURE_DIST <atom1_cid>, <atom2_cid>
SET_PROTON_STATE "none"|"forward"|"reverse"|"shared"
AUDIT_ATOMS [selection=<str>]
EXPORT_RESTRAINTS engine="xray"|"cryoem", out=<path>
```

Example CXL Macro:

```
LOAD "3X2M"
ZOOM_PAIR A:102, A:104, radius=6.5
MAP_FETCH "3X2M", type="2fofc", sigma=1.4
MAP_FETCH "3X2M", type="fofc", sigma=3.0
SET_PROTON_STATE "shared"
MEASURE_DIST //A/102/OD1, //A/104/OD2
EXPORT_RESTRAINTS engine="xray", out="restraints_3x2m.txt"
```

The CXL interpreter in **`carboxmol/scripting/cxl_interpreter.py`** parses commands line-by-line using regular expressions, maps them to internal workbench state methods, and executes them with full error handling and visual telemetry feedback.

8.4 Automated Restraint Injection & Refinement Pipeline

CarboxMol bridges the gap between visualization and physical model refinement:

- User identifies an anomalous carboxyl pair in the 3D viewport.
- User selects **Inject Monoprotonated Restraints**.
- CarboxMol modifies the internal coordinate record, inserting the calculated shared proton atom (*H\delta2* or *H\epsilon2*).
- Generates an external restraint file constraining:
- *d(O_1 - H) = 1.15 \pm 0.03* Å.
- *d(O_2 \cdots H) = 1.30 \pm 0.05* Å.
- Angle *C_1 - O_1 - H = 112^\circ \pm 3^\circ*.
- Dispatches a non-blocking background job to **Carbox Refinement Engine** to execute 10 cycles of restrained refinement.
- Automatically re-loads the refined coordinates and regenerated *2F_o - F_c* and *F_o - F_c* maps, rendering before/after difference comparisons.

CHAPTER SECTION

9. THE MODERN WEB PLATFORM & STUDIO: `webapp/` AND `public/`

The modern Carbox Web Studio (**`webapp/`** mirrored with zero drift to **`public/`**) is an enterprise-grade, browser-based crystallographic workbench built with zero third-party framework overhead (pure Vanilla HTML5, CSS3, and modern ECMAScript).

9.1 Zero-Dependency Vanilla Architecture

By deliberately avoiding heavy frameworks (React, Angular, Vue, Tailwind), Carbox Web Studio achieves:

- Instant loading time (< *150\text{ ms}*).
- Complete immunity from npm dependency rot and breaking build pipeline deprecations.
- Direct, unmediated access to WebGL rendering contexts and high-frequency animation frames (*60\text{ fps}*).
- Complete standalone portability: can be served from GitHub Pages, embedded in a Jupyter notebook, or opened directly from a local **`file://`** URI.

9.2 High-Performance 3D Scene Graph (CarboxMol WebGL Integration)

The 3D rendering pipeline is anchored by **CarboxMol WebGL**, accelerated via custom styling and post-processing routines in **`webapp/studio.js`**:

- **Depth Cueing (Atmospheric Obsidian Fog):** Background atoms fade smoothly into deep slate **`#090d16`**, emphasizing the illuminated active-site pocket.
- **White Focus Contrast Mode:** When a carboxyl dyad is selected, its carbon atoms are automatically illuminated in brilliant alabaster white (**`#ffffff`**), while the surrounding protein skeleton is rendered in subtle semi-transparent CPK or pastel chain colors.
- **Focal Contrast Radius (Carving):** Carbox cuts through massive macromolecular maps by displaying density only within a configurable radius (*4.5* Å pocket, *6.5* Å sphere, or *2.0* Å residue envelope) around the target site.

9.3 Real-Time Density Engine: Mode 0 Decompression & Dual-Mesh Generation

The browser studio loads and renders both primary *2F_o - F_c* maps and differential *\pm F_o - F_c* maps simultaneously:

- Fetches binary maps directly from EBI/PDBe via CORS-enabled streaming ArrayBuffers.
- Evaluates the binary header via **`ensureMode2MapBuffer()`**, transparently decompressing Mode 0 maps into IEEE-754 Float32Array buffers.
- Calculates map RMS directly from CCP4 Word 55 (Float 54), or computes robust standard deviation over non-zero voxels, caching **`volData._cachedRMS`** to eliminate high-frequency recalculation during interactive gestures.
- Generates two distinct isosurface wireframe meshes using the Marching Cubes algorithm:
- ***2F_o - F_c* Map:** Crisp cyan mesh (**`#38bdf8`**), contoured at user-selected *\sigma* (default *1.4\sigma*).
- ***+F_o - F_c* Positive Difference Map:** Emerald green mesh (**`#22c55e`**), contoured at calibrated *+3.0\sigma* to reveal unmodeled features without background noise.
- ***-F_o - F_c* Negative Difference Map:** Coral red mesh (**`#ef4444`**), contoured at calibrated *-3.0\sigma* to highlight steric clashes and over-modeled atoms.

9.4 Precision Trackpad & Mouse Gesture Engine

A primary triumph of Carbox Studio v0.4.5 is its **CarboxMol Precision Gesture Engine**, providing smooth, responsive mouse and trackpad density adjustment.

1. CarboxMol Density Contour Scroll Mode (`D` Shortcut)

- The scroll wheel is natively assigned to adjusting map contour levels (*\sigma*).

• Carbox Studio provides a dedicated toolbar button (**`#btn-vp-scroll-mode`**, with active cyan glow) and a global keyboard shortcut (**`D`** or **`d`**) that toggles scroll wheel behavior between:

• **Density Contour Mode (CarboxMol Mode - Default):** Scrolling adjusts *sigma* level directly.

• **Camera Zoom Mode:** Scrolling zooms the 3D camera.

• If no electron density map is currently active, scrolling automatically falls back to camera zoom, ensuring navigational fluidity.

2. Mac Trackpad Continuous Velocity Normalization

• Standard mouse wheels emit discrete click events (**`deltaY = ±100`** or **`±120`**).

• Mac trackpads emit high-frequency, continuous two-finger drag events (**`deltaY = ±1`** to **`±5`**).

• Carbox Studio distinguishes between discrete clicks and trackpad gestures:

```javascript
let deltaSigma = 0;
if (Math.abs(e.deltaY) >= 50) {
// Discrete mouse wheel click: 0.05σ per notch
deltaSigma = e.deltaY < 0 ? 0.05 : -0.05;
} else {
// Continuous smooth trackpad drag: proportional velocity scaling
deltaSigma = -e.deltaY * 0.0035;
}
```

• Sliding two fingers forward/up increases the contour level (+*sigma*, tighter surface, less noise); sliding backward/down decreases the contour level (-*sigma*, broader envelope).

3. Global Modifier Overrides

• Holding <kbd>Alt</kbd> or <kbd>Ctrl</kbd> while scrolling forces density contour adjustment in any mode.

• Holding <kbd>Shift</kbd> while scrolling adjusts the **Slab Depth Slice** (crystallographic clipping plane), slicing through the protein from *4* Å to *100* Å.

• Pressing <kbd>[</kbd> or <kbd>]</kbd> decrements or increments *sigma* in precise *0.05**sigma* steps.

4. Direct Deck, Slider, and Floating HUD Interaction

• Scrolling the mouse wheel or dragging a trackpad directly over:

• The density control deck (**`#studio-density-deck`**),

• The *2F_o - F_c* slider (**`#studio-density-sigma-range`**),

• The difference slider (**`#studio-density-diff-sigma-range`**),

• The focal contrast card (**`#density-focus-info-card`**), or

• The floating HUD badge (**`#studio-contour-hud`**),

immediately adjusts density contour levels without requiring canvas focus.

• **Click-and-Drag Vertical HUD Adjustment:** Users can click directly on the floating HUD badge and drag vertically up/down to dynamically alter the contour level.

• All visual updates (slider position, numeric readout, floating badge, 3D wireframe mesh) are debounced via **`requestAnimationFrame`**, maintaining an uncompromising *60\text{ fps}*.

```
+-----------------------------------------------------------------------+
|                CARBOX STUDIO GESTURE & SHORTCUT MATRIX                 |
+-----------------------------------+-----------------------------------+
| GESTURE / KEY                     | ACTION                            |
+-----------------------------------+-----------------------------------+
| Scroll Wheel / 2-Finger Trackpad  | Adjust Density Contour (CarboxMol Mode) |
| Alt / Ctrl + Scroll               | Force Density Contour Adjustment  |
| Shift + Scroll                    | Adjust Slab Depth (Clip Planes)   |
| D / d Key                         | Toggle Scroll Mode (Contour / Zoom)|
| [ / ] Keys                        | Step Contour Level (-/+ 0.05σ)    |
| + / - Keys                        | Zoom In / Zoom Out                |
| Space / R Key                     | Reset View & Fit Macromolecule    |
| C Key                             | Recenter Rotation Pivot on Target |
| P / S Key                         | High-Resolution PNG Snapshot      |
| Esc Key                           | Clear Measurement / Close HUD     |
| Direct Drag on HUD Badge          | Vertical Drag Contour Adjustment  |
+-----------------------------------+-----------------------------------+
```

9.5 1D Density Profile Visualizer

Clicking **1D Density Profile** opens an interactive modal calculating the experimental density gradient along the inter-oxygen axis. The visualizer renders:

- The electron density curve *rho(t)* along the vector.
- Peak positions for both carboxylate oxygen atoms.
- The saddle point bridge density.
- Numerical integration of total shared electrons between the oxygens.
- One-click export to high-resolution CSV, SVG, and publication-ready LaTeX **`pgfplots`** tables.

9.6 Interactive Refinement Suite: Carbox Refinement Suite

Carbox Studio embeds a complete refinement configuration suite accessible via the **Launch Refinement Suite** modal:

- **Carbox X-Ray Mode:** Configures maximum-likelihood restrained refinement parameters, twin refinement, jelly-body restraints, and custom weight matrices (*AUTO* vs. user weight).
- **Carbox Cryo-EM Mode:** Configures cryo-EM atomic model refinement or modern X-ray refinement against MTZ/unmerged reflections, with automated hydrogen atom addition in riding or unconstrained positions.
- **CCP4 Cloud Integration:** Generates executable bash scripts and python command-line invocations ready to run locally or upload directly into a CCP4 Cloud project container.

9.7 Trilingual Internationalization Engine (EN, AZ, HE) & RTL Typography

Carbox Studio features complete, native internationalization across three languages:

- **English (en):** The global standard of scientific publishing.
- **Azerbaijani (`az`):** Honoring the native heritage of the authors, Suleiman M Hajizadeh and Garib N Murshudov.
- **Hebrew (he):** Featuring complete bidirectional **Right-to-Left (RTL)** layout mirroring, customized Heebo typography, mirrored flexbox containers, and culturally tailored iconography.

CHAPTER SECTION

10. EXHAUSTIVE DIRECTORY-BY-DIRECTORY MASTER CATALOG

This chapter provides a comprehensive, rigorous catalog of every single directory, subdirectory, and foundational file across the entire Carbox repository.

```
Carbox/
■■■ .agents/                    (Antigravity custom skills & workflow rules)
■■■ carbox/                     (Core Python high-throughput & validation engine)
■■■ carboxmol/                  (Desktop crystallography workbench & CXL bridge)
■   ■■■ bin/                    (Platform-specific wrapper binaries)
■   ■■■ core/                   (Process management & subprocess orchestration)
■   ■■■ deprecated/             (Archived historical legacy modules)
■   ■■■ lib/                    (Monomer library dictionaries & CIF restraints)
■   ■■■ model/                  (Molecular hierarchy & selection algebra)
■   ■■■ qtgui/                  (Qt/PyQt user interface components & dialogs)
■   ■■■ tests/                  (Workbench-specific unit & integration tests)
■   ■■■ view/                   (OpenGL scene graph & camera controllers)
■   ■■■ wrappers/               (CCP4 external tool integration wrappers)
■■■ docs/                       (ReadTheDocs-style trilingual documentation portal)
■■■ examples/                   (Benchmark coordinates, MTZ maps & refined models)
■■■ output/                     (Default execution sink for logs, CSVs & reports)
■■■ public/                     (Production web deployment mirror of webapp/)
■■■ scripts/                    (Automation pipelines, batch processors & builders)
■■■ tests/                      (Complete test suite: 141 pytest modules)
■■■ webapp/                     (Source directory for Carbox Web Studio)
```

10.1 Root Directory Files

• [pyproject.toml](file:///Users/macbookairm2/Documents/GitHub/Carbox/pyproject.toml): Modern PEP-518/PEP-621 build configuration file. Defines project metadata, dependencies (**`gemmi`**, **`numpy`**, **`scipy`**, **`propka`**, **`requests`**, **`pytest`**, **`pytest-cov`**), CLI entrypoints (**`carbox = carbox.cli:main`**), and pytest settings.

• [setup.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/setup.py): Backward-compatible setuptools build script for legacy pip installations.

• [README.md](file:///Users/macbookairm2/Documents/GitHub/Carbox/README.md): Primary repository landing page featuring project description, badges, installation instructions, gallery, and citation information.

• [LICENSE](file:///Users/macbookairm2/Documents/GitHub/Carbox/LICENSE): Full text of the Mozilla Public License 2.0 (MPL-2.0).

• [CARBOX_MASTER_MANUAL.md](file:///Users/macbookairm2/Documents/GitHub/Carbox/CARBOX_MASTER_MANUAL.md): This master monograph.

10.2 `.agents/`

• **Purpose:** Custom configuration, rules, and skill definitions for autonomous AI agents (Google Antigravity IDE ecosystem).

• **Contents:** Workspace customization metadata, pair-programming rules, and automated execution constraints ensuring coding standards and licensing integrity.

10.3 `carbox/` (The Core Engine)

• [__init__.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/__init__.py): Package initializer exposing top-level API functions and version metadata (**`__version__ = "0.4.5"`**).

• [cli.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/cli.py): Complete command-line interface implementation using **`argparse`**. Subcommands include **`scan`**, **`fetch`**, **`audit`**, **`refine`**, and **`batch`**.

• [core.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/core.py): Fundamental geometry engine, atom data classes, KD-tree neighborhood search, and contact detection algorithms.

• [fetch.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/fetch.py): Universal structural data fetcher supporting RCSB PDB, PDBe, AlphaFold DB, and local file portal fallback.

• [graph.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/graph.py): Graph-theoretical clustering and network topology analysis of carboxylate interactions.

• [ligands.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/ligands.py): Small molecule, cofactor, and drug interaction analyzer for carboxylate binding pockets.

• [criteria.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/criteria.py): The Carbox Platform Validation Engine implementing Tukey non-parametric IQR outlier analysis and Fisher Z-score standardization.

• [pka.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/pka.py): Acid dissociation constant (*text{p}K_a*) prediction engine integrating PROPKA and analytical burial calculations.

• [protonation.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/protonation.py): Proton placement geometry and 1D potential energy surface simulation.

• [refinement.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/refinement.py): Automatic restraint generator for Carbox Refinement and Cryo-EM Engines, including **`MODRES`** protonation records.

• [suspicious_atom.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/suspicious_atom.py): Deep atom auditing engine detecting steric clashes, B-factor anomalies, and insertion code inconsistencies.

• [zeta_density.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carbox/zeta_density.py): Gemmi-powered electron density sampler, Fisher Z field computer, and difference peak evaluator.

10.4 `carboxmol/` (The Desktop Crystallographic Workbench)

• [__init__.py](file:///Users/macbookairm2/Documents/GitHub/Carbox/carboxmol/__init__.py): Package initializer exposing workbench components.

• **`core/`**:

• **`CARBOXMOLProcessManager.py`**: Subprocess management, environment variable path configuration, and non-blocking execution of external crystallographic software.

• **`model/`**:

• **`CARBOXMOLModel.py`**: In-memory macromolecular data structure representing models, chains, residues, and atoms.

• **`CARBOXMOLSelectionTree.py`**: Crystallographic atom selection parser and CID interpreter.

• **`view/`**:

• **`CARBOXMOLScene.py`**: 3D OpenGL viewport management, projection matrices, lighting shaders, and clipping slabs.

• **`qtgui/`**:

• **`CARBOXMOLMainWindow.py`**: Primary Qt GUI window and dock layouts.

• **`CARBOXMOLModelWidgets.py`**: Specialized UI tree views, table delegates, and regex-validated input fields.

• **`scripting/`**:

• Native macro and batch execution engine generating Python and CXL scripts for interactive CarboxMol sessions.

• **`wrappers/`**:

• **`adding_stats_to_mmcif_i2/`**: Wrapper script extracting data reduction and refinement statistics from Aimless XML logs and injecting them into mmCIF reflection files.

• **`lib/`**:

• Standard monomer library restraint dictionaries (**`.cif`**) defining ideal geometry for protonated Asp (**`ASH`**) and Glu (**`GLH`**).

• **`deprecated/`**:

• Historical archived modules (e.g., **`carbox_martin/`**) maintained for backward compatibility and provenance tracking.

10.5 `docs/` (Documentation Portal)

• [index.html](file:///Users/macbookairm2/Documents/GitHub/Carbox/docs/index.html): Standalone, light-themed documentation portal covering all aspects of Carbox and CarboxMol.

• [downloads.html](file:///Users/macbookairm2/Documents/GitHub/Carbox/docs/downloads.html): Dedicated downloads hub for standalone binaries, python wheels, and documentation artifacts.

• **`assets/`**: Scalable vector graphics (SVG), logos, and branding assets (**`docs_logo.svg`**, **`download_logo.svg`**).

• **`en/`**, **`az/`**, **`he/`**: Language-specific documentation subportals.

• **`CARBOX_MASTER_MANUAL.md`**: Mirrored master reference manual.

10.6 `examples/`

• Contains benchmark crystallographic structures, MTZ reflection files, and refined models used for verification:

• **`3X2M.pdb`**: Canonical high-resolution benchmark exhibiting catalytic carboxyl dyads.

• **`4RI3.cif`**, **`4RI3-sf.cif`**, **`4RI3_refined.pdb`**, **`4RI3_refined.mtz`**: Full refinement demonstration dataset.

• **`8PO0_refined.pdb`**, **`8PO0_refined.mtz`**: Cryo-EM and high-resolution test structures.

• **`0-carboxmol-history.py`**: Sample session script illustrating integration with CarboxMol.

10.7 `output/`

• Default output sink directory generated by CLI commands, batch scans, and refinement runs. Contains execution logs, CSV anomaly summaries, generated restraint files, and interactive HTML dashboards.

10.8 `public/`

• Production mirror of the **`webapp/`** directory deployed directly to GitHub Pages or static web servers. Maintained at **100% zero drift** relative to **`webapp/`**.

10.9 `scripts/`

• Specialized standalone automation scripts and batch processors:

• **`audit_929_anomalies_criteria.py`**: Executes the complete Carbox Platform audit over the 929 initial PDB anomaly candidates, generating the benchmark decontamination report.

• **`build_docs.py`**: Compiles the trilingual ReadTheDocs-style documentation portal.

• **`run_pdb_pipeline.py`**: Automated daemon for batch-downloading PDB entries, executing Carbox analysis, and aggregating results.

10.10 `tests/`

• Exhaustive automated test suite encompassing **141 passing tests** with 100% pass rate:

• **`test_core.py`**: Core geometry, constants, and basic detection routines.

• **`test_criteria.py`**: Tukey IQR filters, Fisher Z-scores, and anomaly verdicts.

• **`test_protonation_suite.py`**: Proton placement stereochemistry and PES simulations.

• **`test_refinement_engine.py`**: Carbox Refinement restraint generation and **`MODRES`** injection.

• **`test_suspicious_atom.py`**: Multi-criteria atom auditing and clash detection.

• **`test_zeta_density.py`**: Gemmi map sampling and Fisher Z correlation.

• **`test_ligand_interactions.py`**: Small molecule binding site profiling.

• **`test_carbox_graph.py`**: Graph-theoretical network clustering.

• **`test_carboxmol_*.py`**: Full suite testing CarboxMol engine, CXL interpreter, IO, and geometry.

10.11 `webapp/` (Carbox Web Studio)

• [index.html](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/index.html): Primary Dashboard landing page featuring PDB accession input, telemetry cards, and quick-analysis decks.

• [studio.html](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/studio.html): The complete 3D Macromolecular Crystallography Studio interface.

• [downloads.html](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/downloads.html): Downloads hub for users to obtain standalone packages, manuals, and datasets.

• [studio.css](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/studio.css): Deep Obsidian luxury stylesheet defining all desktop docks, viewport controls, and animations.

• [studio.js](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/studio.js): Master Web Studio JavaScript engine (10,800+ lines): 3Dmol.js integration, Mode 0 decompression, CarboxMol precision gesture engine, HUD, and trilingual i18n.

• [app.js](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/app.js): Dashboard controller, telemetry visualizer, and search engine.

• [data.js](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/data.js): Curated pre-indexed database of verified carboxylate dyads.

• [structures.js](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/structures.js): High-throughput structure catalog metadata.

• [sw.js](file:///Users/macbookairm2/Documents/GitHub/Carbox/webapp/sw.js): Progressive Web App (PWA) service worker enabling full offline caching.

• **`assets/`**: High-resolution SVGs, icons, and diagrams.

CHAPTER SECTION

11. LEGAL, LICENSING, COPYRIGHT & OPEN-SOURCE COMPLIANCE

11.1 Mozilla Public License 2.0 (MPL-2.0) Architecture

The entire Carbox & CarboxMol project is licensed under the **Mozilla Public License, Version 2.0 (MPL-2.0)**.

Key Principles of MPL-2.0:

- **Commercial & Academic Freedom:** Any individual, academic institution, or commercial entity is free to use, modify, distribute, or run the software without payment of royalties.
- **File-Level Copyleft:** Any modifications made to existing files covered by MPL-2.0 must be distributed under the MPL-2.0 license if made available to external parties.
- **Larger Works:** The MPL-2.0 explicitly permits combining Carbox code with proprietary or differently-licensed software into a "Larger Work," provided the MPL-covered files remain open-source under MPL-2.0.
- **Patent Grant:** Contributors grant a non-exclusive, worldwide, royalty-free patent license covering patent claims necessarily infringed by their contributions.

11.2 Dual Authorship Attribution & Academic Moral Rights

In accordance with international academic integrity standards and intellectual property conventions, copyright and authorship are universally attributed to:

- **Suleiman M Hajizadeh** — Concept, Lead Software Architecture, Core Algorithm Engineering, Web Studio Engine.
- **Garib N Murshudov** — Scientific Directorship, Crystallographic Validation Philosophy, Statistical Formulation, Senior Advisorship.

All academic papers, theses, posters, software distributions, or commercial products utilizing the Carbox codebase or its derived findings must include explicit attribution:

Key Scientific Insight

Suleiman M Hajizadeh & Garib N Murshudov (2026). Carbox & CarboxMol: Computational Framework for High-Throughput Detection, Carbox Validation, and Refinement of Anomalous Carboxylate Microenvironments in Macromolecular Crystallography. Version 0.4.5.

11.3 Open-Source Crystallographic Ecosystem & Standards

Carbox and CarboxMol operate upon and harmonize with open-source structural biology libraries and standards:

- **Macromolecular Crystallographic Formats:** High-precision parsing and handling of PDB, mmCIF, MTZ, and CCP4 map standards.
- **Computational Libraries:** Integration with high-performance numerical and structural biology modules (Gemmi, 3Dmol.js).
- **Refinement Interoperability:** Carbox X-Ray and Cryo-EM refinement parameter generation and model validation.

11.4 Statement of Non-Infringement & Ethical Integrity

A thorough legal and intellectual property audit confirms that:

- **Zero Infringing Material:** The Carbox repository contains no pirated software, cracked binaries, confidential industrial trade secrets, or unauthorized proprietary code.
- **Open Science Standards:** All benchmark datasets (**`3X2M`**, **`4RI3`**, **`8PO0`**) are derived from publicly accessible Protein Data Bank records under Creative Commons Zero (CC0) public domain dedications.
- **Code Provenance:** All Python, JavaScript, HTML, and CSS source code was written clean-room by the authors or adapted from permissively licensed open-source projects with explicit attribution retained in the source headers.

CHAPTER SECTION

12. BIBLIOGRAPHY & COMPREHENSIVE SCIENTIFIC REFERENCES

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End of Carbox & CarboxMol Definitive Master Manual (Release 0.4.5)

APPENDIX A

The Validated Golden LBHB Benchmark Dataset (All 67 Sites)

The following multi-page catalog presents all 67 validated, high-confidence Low-Barrier Hydrogen Bond (LBHB) sites mined from sub-angstrom and atomic-resolution crystal structures ($d_{\text{res}} \le 2.0\text{ \AA}$) that successfully passed all four stages of the Carbox Platform Validation Engine. Every site exhibits atomic displacement parameters well within the structure core distribution ($B \le Q_3 + 1.5\cdot IQR$), absence of negative difference electron density, full occupancy (1.0), and severe dielectric water burial ($d_w \ge 3.0\text{ \AA}$):

APPENDIX B

Decontamination Audit: Rejected Artifacts Catalog

Below is a representative catalog of false-positive carboxyl 'anomalies' rejected by the Carbox Platform Validation Engine during the high-throughput audit of 929 apparent contacts. Specific failure modes—thermal disorder, steric clashes, alternate conformation splits, and unmodeled divalent cations—are documented to provide structural biologists with diagnostic insight into common refinement artifacts:

APPENDIX C

Automated Test Suite Execution Log (141 Unit & Integration Tests)

Carbox maintains a comprehensive automated testing infrastructure executed via pytest. All 141 tests pass with a 100% pass rate in 4.39 seconds on Apple Silicon M2 hardware. Below is the complete execution register documenting test module, test name, specific biochemical functionality validated, and verification status:

Automated Test Suite Execution Log (Part 1 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
`test_carbox_graph.py`	**test_molecular_graph_basic_operations**	Validates node/edge insertion, adjacency, and degree counts. **[PASSED]**
`test_carbox_graph.py`	**test_directed_molecular_graph**	Tests directed graph topologies and edge flow directions. **[PASSED]**
`test_carbox_graph.py`	**test_traversal_bfs_dfs**	Verifies Breadth-First and Depth-First search tree correctness. **[PASSED]**
`test_carbox_graph.py`	**test_dijkstra_shortest_paths**	Validates Dijkstra shortest path distances between distant residues. **[PASSED]**
`test_carbox_graph.py`	**test_all_simple_paths**	Tests exhaustive simple path discovery between carboxyl nodes. **[PASSED]**
`test_carbox_graph.py`	**test_connected_components**	Identifies isolated carboxyl hydrogen-bond networks in protein cores. **[PASSED]**
`test_carbox_graph.py`	**test_cycle_basis_carboxyl_rings**	Computes fundamental cycle basis for carboxylate chelate rings. **[PASSED]**
`test_carbox_graph.py`	**test_centrality_betweenness_closeness**	Computes betweenness and closeness centrality for key dyad nodes. **[PASSED]**
`test_carbox_graph.py`	**test_fruchterman_reingold_layout_2d**	Executes 2D force-directed spring simulation with zero external deps. **[PASSED]**
`test_carboxmol_geometry_suite.py`	**test_crystallographic_unit_cell_and_symmetry**	Validates space group symmetry operations and fractionalization. **[PASSED]**
`test_carboxmol_ideal_refinement.py`	**test_bond_energy_and_gradients**	Verifies harmonic bond stretch energy and analytical force gradients. **[PASSED]**
`test_carboxmol_ideal_refinement.py`	**test_valence_angle_energy_and_gradients**	Verifies valence angle deformation energy and analytical gradients. **[PASSED]**
`test_carboxmol_ideal_refinement.py`	**test_chiral_volume_calculation**	Computes signed chiral volumes to preserve C-alpha stereochemistry. **[PASSED]**
`test_carboxmol_ideal_refinement.py`	**test_real_space_carboxyl_minimizer**	Executes real-space L-BFGS minimizer for carboxyl hydrogen placement. **[PASSED]**
`test_carboxmol_io_suite.py`	**test_pdb_io_parse_and_write**	PDB parser and writer with round-trip coordinate invariance. **[PASSED]**
`test_carboxmol_io_suite.py`	**test_cif_io_parse_and_write**	mmCIF dictionary-based parser supporting loop_ categories. **[PASSED]**
`test_carboxmol_io_suite.py`	**test_cxl_io_formatting_and_parsing**	Carbox eXtensible Language serialization and AST parser. **[PASSED]**
`test_carboxmol_io_suite.py`	**test_map_io_ccp4_header**	Validates binary CCP4 map header parsing (Word 55 RMS, grid spacing). **[PASSED]**

Automated Test Suite Execution Log (Part 2 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
test_carboxmol_macro.py	**test_parse_cxl_script**	Parses complex multi-statement CXL macro scripts into execution plans. **[PASSED]**
test_carboxmol_macro.py	**test_validate_cxl_script_valid**	Validates syntax integrity for standard crystallographic CXL macros. **[PASSED]**
test_carboxmol_macro.py	**test_validate_cxl_script_with_warnings**	Detects syntax warnings and deprecated CXL selection tokens. **[PASSED]**
test_carboxmol_metal_suite.py	**test_metal_detector_and_polyhedra**	Detects Mg, Ca, Zn coordination polyhedra and coordination numbers. **[PASSED]**
test_carboxmol_metal_suite.py	**test_carboxylate_metal_bridging**	Identifies bridging carboxylates coordinating shared divalent cations. **[PASSED]**
test_carboxmol_pli_suite.py	**test_salt_bridge_detection**	Identifies Arg/Lys guanidinium/ammonium to Asp/Glu carboxylate bridges. **[PASSED]**
test_carboxmol_pli_suite.py	**test_pi_stacking_detection**	Identifies aromatic pi-pi stacking interactions near catalytic dyads. **[PASSED]**
test_carboxmol_pli_suite.py	**test_hbond_finder_lbhb**	Dedicated LBHB finder identifying ultra-short contacts (d <= 2.55 A). **[PASSED]**
test_carboxmol_pli_suite.py	**test_cbeta_deviation_validator**	Validates C-beta atomic deviations to prevent distorted rotamers. **[PASSED]**
test_carboxmol_pli_suite.py	**test_density_isosurface_extraction**	Marching cubes isosurface triangulation from 3D volumetric grids. **[PASSED]**
test_carboxmol_restraints_refine.py	**test_external_restraints_generation**	Generates Carbox external distance restraints with sigma weighting. **[PASSED]**
test_carboxmol_restraints_refine.py	**test_modres_generation_and_injection**	Injects MODRES records for neutral ASH/GLH residues into PDB/CIF. **[PASSED]**
test_carboxmol_restraints_refine.py	**test_lbhb_restraint**	Generates single-well symmetric LBHB restraints for ultra-short pairs. **[PASSED]**
test_carboxmol_restraints_refine.py	**test_covalent_link_cif**	Writes mmCIF chem_comp_link records for proton-sharing dyads. **[PASSED]**
test_carboxmol_restraints_refine.py	**test_refinement_pipeline_and_adapters**	Executes mock refinement pipeline and parses refinement stats. **[PASSED]**
test_carboxmol_scripting_bridge.py	**test_cxl_interpreter**	Evaluates runtime variables and arithmetic expressions in CXL. **[PASSED]**
test_carboxmol_scripting_bridge.py	**test_macro_generator_crystallographic**	Generates automated CarboxMol macro and inspection script files. **[PASSED]**
test_carboxmol_scripting_bridge.py	**test_macro_generator_scene**	Serializes 3D viewport orientation, zoom, and clipping planes. **[PASSED]**

Automated Test Suite Execution Log (Part 3 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
`test_carboxmol_scripting_bridge.py`	**test_html_report_generator**	Outputs self-contained HTML reports with embedded CarboxMol WebGL viewers. **[PASSED]**
`test_core.py`	**test_carboxyl_oxygens_keys**	Validates Asp (OD1, OD2) and Glu (OE1, OE2) carboxyl atom mappings. **[PASSED]**
`test_core.py`	**test_carboxyl_oxygens_tuples**	Ensures immutable tuple lookups for carboxylate oxygen identifiers. **[PASSED]**
`test_core.py`	**test_lbhb_cutoff_below_max**	Guarantees LBHB threshold (2.55 A) is strictly below general cutoff (3.20 A). **[PASSED]**
`test_core.py`	**test_water_burial_cutoff_positive**	Validates first-hydration shell water burial distance (3.00 A). **[PASSED]**
`test_core.py`	**test_water_names_present**	Ensures HOH, WAT, DOD, D2O are all indexed as crystallographic waters. **[PASSED]**
`test_core.py`	**test_buried_lbhb**	Classifies buried ultra-short contact as Likely Protonated - Buried LBHB. **[PASSED]**
`test_core.py`	**test_buried_short_hbond**	Classifies buried 2.55-3.00 A contact as Buried Short H-Bond. **[PASSED]**
`test_core.py`	**test_solvent_exposed_short_hbond**	Classifies water-exposed contact as Possibly Protonated - Solvent Exposed. **[PASSED]**
`test_core.py`	**test_solvent_exposed_ultra_short**	Audits ultra-short solvent-exposed contact for dielectric shielding. **[PASSED]**
`test_core.py`	**test_lbhb_cutoff_value**	Verifies LBHB distance boundary constant equals 2.55 A. **[PASSED]**
`test_core.py`	**test_max_oo_distance_value**	Verifies general carboxyl screening cutoff constant equals 3.20 A. **[PASSED]**
`test_core.py`	**test_cli_import_and_main**	Tests CLI argument parsing, help output, and exit code handling. **[PASSED]**
`test_core.py`	**test_carboxmol_script_generator**	Validates generation of desktop CarboxMol inspection scripts. **[PASSED]**
`test_core.py`	**test_carboxmol_view_generator**	Validates generation of interactive 3D visual scenes. **[PASSED]**
`test_core.py`	**test_refmac_instructions**	Tests generation of Carbox refinement keyword command decks. **[PASSED]**
`test_core.py`	**test_servalcat_adapter**	Tests Carbox parameter file formatting and residue protonation. **[PASSED]**
`test_core.py`	**test_standalone_html_generator**	Builds complete, zero-dependency standalone HTML 3D visualization. **[PASSED]**

Automated Test Suite Execution Log (Part 4 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
`test_core.py`	**test_density_sampling_profile**	Extracts 1D electron density line profile along donor-acceptor vector. **[PASSED]**
`test_core.py`	**test_fo_fc_difference_peak_evaluation**	Evaluates positive and negative difference density peaks. **[PASSED]**
`test_cxl_and_metal.py`	**test_generate_cxl_script**	Generates full CXL macro scripts from identified carboxyl contacts. **[PASSED]**
`test_cxl_and_metal.py`	**test_detect_metal_coordination_bridging**	Detects bridging divalent metal coordination between two carboxylates. **[PASSED]**
`test_cxl_and_metal.py`	**test_detect_metal_coordination_no_metal**	Confirms non-metal organic hydrogen bonds return negative for metals. **[PASSED]**
`test_fetch_universal.py`	**test_fetch_universal_local_portal**	Fetches structures from local cache directory with high priority. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_rcsb_pdb**	Builds correct HTTPS download URL for RCSB PDB coordinate files. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_rcsb_cif**	Builds correct HTTPS download URL for RCSB mmCIF files. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_rcsb_bcif**	Builds BinaryCIF URL for high-speed network streaming. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_fasta**	Builds FASTA sequence query URL for macromolecular chains. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_xml**	Builds PDB XML metadata query URL. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_alphafold**	Builds AlphaFold DB model download URL from UniProt accession IDs. **[PASSED]**
`test_fetch_universal.py`	**test_get_fetch_url_pdbe**	Builds PDBe European Bioinformatics Institute download URLs. **[PASSED]**
`test_fetch_universal.py`	**test_fetch_convenience_functions**	Tests unified fetch wrappers across multiple repository endpoints. **[PASSED]**
`test_gemmi_density.py`	**test_gemmi_installation**	Validates C++/Python Gemmi bindings and version availability. **[PASSED]**
`test_gemmi_density.py`	**test_fetch_and_parse_ccp4_map**	Parses binary CCP4 electron density grids using Gemmi. **[PASSED]**
`test_gemmi_density.py`	**test_sample_atomic_densities_all_atoms**	Samples real-space 2Fo-Fc electron density at all atomic coordinates. **[PASSED]**
`test_gemmi_density.py`	**test_get_protein_density_summary**	Computes mean, sigma, and skewness of macromolecular density values. **[PASSED]**

Automated Test Suite Execution Log (Part 5 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
test_ligand_interactions.py	**test_extract_all_ligands_5rcg**	Extracts all small-molecule drug/ligand entities from PDB 5RCG. **[PASSED]**
test_ligand_interactions.py	**test_analyze_comprehensive_ligand_interactions_reg**	Screens ligand carboxylates interacting with active site Asp/Glu. **[PASSED]**
test_ligand_interactions.py	**test_analyze_ligand_carboxylates_backward_compatibility**	Ensures legacy API compatibility for ligand screening functions. **[PASSED]**
test_metal_and_proton_features.py	**test_proton_transfer_forward_and_reverse**	Simulates forward and reverse ('■ks a■■z') proton trajectories. **[PASSED]**
test_metal_and_proton_features.py	**test_proton_transfer_bidirectional_roundtrip**	Validates thermodynamic path conservation in roundtrip simulations. **[PASSED]**
test_metal_and_proton_features.py	**test_metal_coordination_site_detection**	Detects transition metal coordination geometry and coordination numbers. **[PASSED]**
test_metal_and_proton_features.py	**test_metal_color_palettes**	Validates standard CPK and bioinorganic metal visualization palettes. **[PASSED]**
test_criteria.py	**test_b_statistics_computation**	Computes non-parametric Tukey quartiles (Q1, Median, Q3, IQR) over B-factors. **[PASSED]**
test_criteria.py	**test_fisher_transformation**	Applies standardized Fisher log-transformation Z = (ln B - mean) / sd. **[PASSED]**
test_criteria.py	**test_evaluate_dyad_golden**	Validates canonical atomic-resolution LBHB dyad as GOLDEN_RELIABLE. **[PASSED]**
test_criteria.py	**test_evaluate_dyad_extreme_outlier**	Rejects extreme thermal outlier contact (B > Q3 + 3*IQR). **[PASSED]**
test_criteria.py	**test_evaluate_dyad_suspicious_low_b**	Flags suspicious low B-factors indicating coordinate pinning artifacts. **[PASSED]**
test_pdb_pipeline_integration.py	**test_manifest_query_small**	Queries RCSB Search API v2 with resolution and method constraints. **[PASSED]**
test_pdb_pipeline_integration.py	**test_parse_and_scan_cached_structure**	End-to-end mmCIF parsing, contact screening, and verdict assignment. **[PASSED]**
test_pdb_pipeline_integration.py	**test_markdown_report_generation**	Builds comprehensive scientific Markdown reports from survey runs. **[PASSED]**
test_pka_analyzer.py	**test_run_propka_on_sample_pdb**	Executes PROPKA 3 on sample PDB files and parses pKa output. **[PASSED]**
test_pka_analyzer.py	**test_detect_anomalous_carboxyl_pka_filtering**	Identifies Asp/Glu residues with extreme pKa shifts (|Delta-pKa| > 3.0). **[PASSED]**
test_pka_analyzer.py	**test_missing_file_handling**	Validates robust exception handling when coordinate files are missing. **[PASSED]**

Automated Test Suite Execution Log (Part 6 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
test_protonation_suite.py	**test_library_exports**	Verifies all required protonation classes and functions are exported. **[PASSED]**
test_protonation_suite.py	**test_mismodeled_electrostatic_clash**	Detects unprotonated carboxyl pairs modeled in steric clash. **[PASSED]**
test_protonation_suite.py	**test_correct_lbhb_monoprotonated_model**	Builds mono-protonated LBHB model with shared hydrogen geometry. **[PASSED]**
test_protonation_suite.py	**test_difference_density_validation**	Simulates omit difference density maps around the shared proton site. **[PASSED]**
test_protonation_suite.py	**test_simulate_proton_transfer_ultra_short_lbhb**	Calculates collapse of central barrier into single symmetric well. **[PASSED]**
test_protonation_suite.py	**test_simulate_proton_transfer_standard_double_well**	Calculates standard asymmetric double-well PES for d = 2.80 A. **[PASSED]**
test_refinement_engine.py	**test_servalcat_is_available_and_version**	Checks Carbox installation and version query compatibility. **[PASSED]**
test_refinement_engine.py	**test_generate_external_restraints**	Generates Carbox external distance restraints with user-defined sigma. **[PASSED]**
test_refinement_engine.py	**test_inject_protonation_modres_pdb**	Injects MODRES records for neutral residues into PDB formatted models. **[PASSED]**
test_refinement_engine.py	**test_inject_protonation_modres_cif**	Injects MODRES records into mmCIF dictionary categories. **[PASSED]**
test_refinement_engine.py	**test_generate_xtal_refine_script**	Generates crystallographic refinement shell scripts for Carbox. **[PASSED]**
test_refinement_engine.py	**test_generate_spa_refine_script**	Generates single-particle cryo-EM refinement scripts for Carbox Cryo-EM. **[PASSED]**
test_refinement_engine.py	**test_generate_fofc_omit_script**	Builds omit difference map calculation scripts for validating hydrogen. **[PASSED]**
test_refinement_engine.py	**test_generate_h_add_script**	Generates scripts for adding explicit riding hydrogens. **[PASSED]**
test_refinement_engine.py	**test_servalcat_adapter_backward_compatibility**	Ensures Carbox adapter parameters remain supported. **[PASSED]**
test_suspicious_atom.py	**test_carbox_library_exports**	Verifies suspicious atom auditing functions are exposed in carbox API. **[PASSED]**
test_suspicious_atom.py	**test_lbhb_suspicious_atom_detection**	Audits LBHB oxygen atoms against thermal and clash criteria. **[PASSED]**
test_suspicious_atom.py	**test_normal_atom_audit**	Confirms standard well-ordered core atoms pass all audit criteria. **[PASSED]**

Automated Test Suite Execution Log (Part 7 of 7)

Test Module	Test Case Name	Functionality Validated & Pass Status
test_suspicious_atom.py	**test_bfactor_anomaly_audit**	Detects anomalously high or pinned B-factors in carboxyl sidechains. **[PASSED]**
test_suspicious_atom.py	**test_steric_clash_detection**	Calculates van der Waals overlaps between non-bonded heavy atoms. **[PASSED]**
test_suspicious_atom.py	**test_gemmi_compatibility**	Ensures complete compatibility with native Gemmi Atom and Residue objects. **[PASSED]**
test_suspicious_atom.py	**test_insertion_code_and_restraint_export**	Handles PDB insertion codes (e.g., Asp100A) in restraint output. **[PASSED]**
test_suspicious_atom.py	**test_metal_coordination_object_support**	Validates coordination checking with metal coordination objects. **[PASSED]**
test_zeta_density.py	**test_valid_accessions**	Validates alphanumeric PDB accession codes (e.g. 3X2M, 5SAO). **[PASSED]**
test_zeta_density.py	**test_invalid_path_traversal_accessions**	Rejects malicious path traversal strings (e.g., ../../etc/passwd). **[PASSED]**
test_zeta_density.py	**test_fisher_z_field_computation_positive**	Validates Fisher Z-field calculation for correlated density. **[PASSED]**
test_zeta_density.py	**test_fisher_z_field_computation_inverted**	Validates Fisher Z-field calculation for anti-correlated density. **[PASSED]**
test_zeta_density.py	**test_calculate_all_zeta_scores**	Calculates standardized zeta scores across full crystallographic unit cells. **[PASSED]**
test_zeta_density.py	**test_evaluate_metal_zeta_environment**	Evaluates density signal-to-noise ratio around metal coordination sites. **[PASSED]**

APPENDIX D

Comprehensive API Reference for carbox.* Core Modules

This appendix provides a developer-level API reference for the primary classes, methods, and functions exported by the 11 modules of the core carbox Python library:

Class: carbox.core.ContactFinder

High-throughput spatial geometric contact mining engine.

- **`find_contacts(structure: gemmi.Structure, max_distance: float = 3.20) -> List[CarboxylPair]`**
  Builds a KD-Tree over all carboxylate oxygens (OD1, OD2 in Asp; OE1, OE2 in Glu) and extracts all candidate pairs satisfying d(O...O) <= max_distance.
- **`calculate_water_burial(pair: CarboxylPair, waters: List[gemmi.Atom]) -> float`**
  Computes the minimum Euclidean distance d_w between either carboxyl oxygen and the closest crystallographic water atom.
- **`classify_verdict(d_oo: float, d_w: float) -> str`**
  Assigns deterministic protonation state: Buried LBHB (d <= 2.55, d_w >= 3.0), Buried Short H-Bond, or Solvent Exposed.

Class: carbox.graph.MolecularGraph

Zero-dependency graph engine implementing molecular topology analysis.

- **`add_node(node_id: str, **attributes) -> None`**
  Inserts an atomic or residue node with chemical metadata.
- **`add_edge(u: str, v: str, weight: float = 1.0, **attributes) -> None`**
  Inserts a contact or covalent edge with distance weighting.
- **`dijkstra_shortest_path(source: str, target: str) -> Tuple[List[str], float]`**
  Computes optimal shortest path through hydrogen-bond networks.
- **`find_cycle_basis() -> List[List[str]]`**
  Computes cycle basis for metal chelate and carboxyl ring structures.
- **`fruchterman_reingold_layout(iterations: int = 50) -> Dict[str, Tuple[float, float]]`**
  Simulates 2D force-directed spring layout without NetworkX.

Class: carbox.validation.criteria.DyadValidationEngine

Crystallographic validation engine implementing Carbox criteria.

- **`compute_b_statistics(b_factors: List[float]) -> BFactorStatistics`**
  Calculates Q1, Median, Q3, IQR, and Tukey mild/extreme outlier fences.
- **`compute_fisher_z(b_factor: float, stats: BFactorStatistics) -> float`**
  Computes standardized log-normal Fisher Z-score: Z = (ln B - mean_ln) / sd_ln.
- **`evaluate_dyad(pair: CarboxylPair, stats: BFactorStatistics) -> DyadValidationResult`**
  Audits candidate against thermal fences, negative density, and occupancy.

Class: carbox.protonation.ProtonationSimulator

Quantum Morse potential energy surface and proton transfer trajectory simulator.

- **`build_morse_pes(d_oo: float, delta_pka: float = 0.0) -> PESProfile`**
  Calculates 1D double Morse potential along the donor-acceptor coordinate.
- **`solve_ground_state(pes: PESProfile) -> WavefunctionResult`**
  Solves 1D Schrödinger equation to evaluate proton tunneling and ZPE barrier crossing.

- **`compute_proton_transfer_forward_and_reverse(pair: CarboxylPair) -> TrajectoryResult`**
  Simulates forward and reverse ('■ks a■■z') proton trajectories.

Class: carbox.refine.restraints.RefinementRestraintEngine

Automated restraint generator for Carbox crystallographic refinement engines.

- **`generate_refinement_external_restraint(pair: CarboxylPair, sigma: float = 0.02) -> str`**
  Outputs external distance directives maintaining LBHB geometry.
- **`generate_cryo_parameters(pair: CarboxylPair) -> Dict[str, Any]`**
  Outputs Cryo-EM parameters specifying explicit neutral ASH/GLH residues.
- **`inject_modres_records(model_path: Path, output_path: Path, residues: List[str]) -> None`**
  Injects standard MODRES records into PDB/mmCIF headers.

Class: carbox.suspicious_atom.SuspiciousAtomAuditor

Multi-criteria atomic validation and steric clash detector.

- **`audit_atom(atom: gemmi.Atom, neighbors: List[gemmi.Atom]) -> AuditVerdict`**
  Evaluates steric overlap, B-factor outliers, and partial occupancy.
- **`detect_steric_clashes(pair: CarboxylPair, vdw_tolerance: float = 0.40) -> bool`**
  Identifies unphysical van der Waals interpenetration between carboxylates.

Class: carbox.zeta_density.ZetaDensitySampler

Gemmi volumetric map density sampler and local signal-to-noise evaluator.

- **`sample_atomic_densities(map_grid: gemmi.FloatGrid, atoms: List[gemmi.Atom]) -> List[float]`**
  Performs trilinear interpolation of electron density at atomic coordinates.
- **`compute_fisher_z_field(density_grid: gemmi.FloatGrid) -> gemmi.FloatGrid`**
  Computes standardized Fisher Z-field across full crystallographic unit cells.

APPENDIX E

Restraint and Parameterization Guide for Carbox Refinement

Modern macromolecular refinement suites enforce standard stereochemical dictionary targets that assume unprotonated carboxylate oxygens cannot approach closer than $2.80–3.00\text{ \AA}$. When refining atomic-resolution crystal or cryo-EM structures containing authentic Low-Barrier Hydrogen Bonds, custom restraints must be supplied to prevent the refinement program from artificially pulling the oxygens apart.

E.1 Carbox External Distance Restraints Syntax

Carbox supports external distance restraints via the exte dist keyword directive. Carbox generates these directives with standard uncertainty $\sigma = 0.02\text{ \AA}$:

```
# Carbox External Distance Restraints for Validated LBHB Dyad
# Authors: Suleiman M Hajizadeh & Garib N Murshudov
# Syntax: exte dist first chain <C1> resi <R1> atom <A1> second chain <C2> resi <R2> atom <A2> value <VAL>
sigma <SIG>
exte dist first chain A resi 149 atom OD2 second chain A resi 162 atom OE1 value 2.512 sigma 0.02
exte dist first chain A resi 35  atom OD2 second chain A resi 47  atom OD1 value 2.480 sigma 0.02
exte dist first chain A resi 95  atom OE1 second chain A resi 112 atom OE2 value 2.620 sigma 0.03
```

E.2 Carbox Neutral Residue Refinement Invocation

Carbox accepts explicit neutral amino acid parameterizations (ASH for protonated Aspartate, GLH for protonated Glutamate). The following command line executes atomic refinement while treating Asp149 as a neutral ASH residue with riding hydrogens:

```
# Carbox Atomic Refinement with Explicit Neutral Carboxyl Residues
carbox refine \
  --model input.cif \
  --h-mode all \
  --protonate-residues "A:149:ASH,A:162:GLU" \
  --hklin structure_factors.mtz \
  --cycles 10 \
  --output-prefix refined_carbox_model
```

E.3 CarboxMol Python / CXL Macro for Real-Space Inspection

```
# CarboxMol Inspection Macro (Python API)
import carboxmol
session = carboxmol.create_session()
model = session.open_structure('3x2m.cif')
density = session.open_density_map('3x2m.mtz')
session.set_rotation_center(12.45, 34.12, -8.76)
session.set_zoom(25.0)
density.set_contour_level(1.2)
# Add distance object across LBHB
session.measure_distance(model, 'A', 149, '', 'OD2', 'A', 162, '', 'OE1')
```

APPENDIX F

Comprehensive Scientific Bibliography & Primary References

The following primary scientific literature forms the theoretical, physical, and crystallographic foundation of the Carbox & CarboxMol platform:

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